

Robust Individual and Group Fair Classification

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Abstract

Fair classification algorithms ensure equitable treatment across protected groups and similar individuals, but their performance degrades when training data is corrupted. We study fair binary classification under an α -corruption model where features, labels, and protected attributes may be arbitrarily corrupted. We first analyze a Naive-FAIR approach enforcing joint demographic parity (DP) and metric individual fairness (IF) constraints on corrupted data, deriving bounds showing clean-distribution violations are bounded by $\frac{\alpha}{(1-\alpha)\pi_0+\alpha}$ for DP (with π_0 the minority group proportion) and $2\alpha - \alpha^2$ for IF. To mitigate corruption, we propose DRO-FAIR (Distributionally Robust Optimization for Fairness), which minimizes worst-case fairness violations over uncertainty sets defined via total variation distance. The radii of these sets are calibrated to the level of corruption, ensuring that the clean distribution lies within the uncertainty set without requiring identification of corrupted samples. We prove DRO-FAIR achieves robustness guarantees that provide fairness certificates on the clean distribution without statistical rate degradation, maintaining $O(\sqrt{D_{VC} \log n/n})$ generalization. Experiments on three real-world benchmarks show DRO-FAIR consistently reduces fairness violations across corruption levels ($\alpha \in \{0.1, 0.2, 0.3\}$), with moderate accuracy trade-offs (1 to 4%) and $12\times$ computational overhead.

1. Introduction

Machine learning systems deployed in high-stakes niches such as healthcare (Obermeyer et al., 2019), criminal justice (Dressel & Farid, 2018), and facial recognition (Buolamwini & Gebru, 2018) can inherit and amplify societal biases. Algo-

rithmic fairness studies these biases and develops mitigation strategies, broadly centered on two complementary yet often competing notions: group fairness (Calders et al., 2009) and individual fairness (Dwork et al., 2012).

Group fairness is commonly formalized as demographic parity (DP), which requires parity of outcomes across pre-defined demographic groups (gender, race, etc.). In contrast, individual fairness (IF) mandates that similar individuals receive similar predictions, as measured by a task-specific similarity metric. Notably, enforcing DP can cause severe individual level unfairness by assigning different outcomes to otherwise similar individuals (Binns, 2020). This tension highlights the need for learning frameworks that account for both group-level and individual-level fairness considerations.

A substantial body of work has proposed in-processing methods to enforce DP (Zafar et al., 2019; Agarwal et al., 2018; Lokhande et al., 2020; Tran et al., 2021; Cousins, 2022; Cotter et al., 2019), typically by augmenting training objectives with fairness-aware constraints or regularizers and establishing generalization or sample complexity guarantees. Complementary work has analyzed the sample complexity of metric-based notions of IF (Rothblum & Yona, 2018). However, a common implicit assumption in existing methods is that the training data, features, labels, and demographic groups, are clean and reliably observed.

As fairness constraints are increasingly enforced on large, real-world datasets, violations of this clean-data assumption become unavoidable. Such corruption encompasses a range of issues, from random label noise to adversarial feature manipulations. Recent empirical studies demonstrate that enforcing fairness constraints on corrupted data, whether through label noise (Wang et al., 2021) or more general corruption (Konstantinov & Lampert, 2022), often results in degraded fairness guarantees on the clean distribution. For instance, consider a credit default prediction task where two applicants of different races have similar true creditworthiness. Due to measurement error in financial features, such as incomplete income records or noisy credit histories, one applicant from a protected racial group is observed with distorted features. A classifier trained to satisfy DP on this corrupted data may assign different credit decisions to these otherwise similar individuals, thereby violating IF.

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

Moreover, when corruption systematically affects feature quality by group, such a classifier may fail to achieve DP on the clean population due to distorted group-conditional statistics.

These limitations motivate approaches that are robust to data corruption while providing fairness guarantees. Distributionally Robust Optimization (DRO) (Shapiro et al., 2014; Rahimian & Mehrotra, 2022) offers a principled framework for minimizing worst-case risk over uncertainty sets around the empirical distribution. Prior work has applied DRO to group fairness (Hashimoto et al., 2018; Sagawa et al., 2020; Taskesen et al., 2020), primarily to address natural distribution shifts across demographic groups. However, existing DRO-based fairness approaches focus almost exclusively on group-level objectives and do not address IF, nor do they account for simultaneous corruption of features, labels, and protected attributes.

Our Contributions. We present a unified framework for fair binary classification under data corruption, providing the first provable guarantees for both Demographic Parity (DP) and Individual Fairness (IF) under the α -corruption model (Valiant, 1985; Kearns & Li, 1993). Our main contributions are:

- **Tight Corruption Bounds.** We prove that corruption shifts each group’s acceptance rate by $\frac{\alpha}{(1-\alpha)\pi_j + \alpha}$ (with π_j the group proportion), revealing disproportionate harm to minority groups. For IF, the shift is $2\alpha - \alpha^2$ (Theorems 4.2–4.3; Corollaries 4.4–4.5). Our finite-sample analysis reveals sample requirements scaling with $1/\alpha$ (corruption) and $D_{VC}/[(1-\alpha)\pi_0]$ (complexity-minority interaction), confirming the naive approach incurs these penalties (Theorem 5.1).
- **DRO-FAIR: Robust Fairness with Provable Guarantees.** We propose DRO-FAIR, which enforces fairness over corruption-calibrated uncertainty sets with tight radii derived from our bounds (Remark 6.2). Unlike the naive approach, DRO-FAIR guarantees any feasible classifier satisfies the fairness constraints on the *clean* distribution, while maintaining the same statistical rates as non-robust methods (Theorem 6.1).
- **Empirical Validation.** We provide a tractable min-max algorithm (Algorithm 1) and validate on Adult, Credit, and LSAC. Under moderate corruption ($\alpha \leq 0.3$), DRO-FAIR achieves up to 83% lower DP violations and 3 to $5\times$ lower IF violations across datasets, with 1 to 4% accuracy cost, demonstrating practical robustness for real-world deployment.

2. Related Work

A significant line of work studies fairness under natural distribution shift (NS), where data is uncorrupted but the test distribution differs across subpopulations or environments. Under the PAC model, Rothblum & Yona (2018) establish polynomial sample complexity bounds for metric-based individual fairness assuming i.i.d. samples from a clean distribution. DRO (Shapiro et al., 2014; Rahimian & Mehrotra, 2022) has been widely adopted in this setting, with uncertainty sets defined via f -divergences (Namkoong & Duchi, 2016) or Wasserstein distances (Sinha et al., 2018). In the fairness context, group DRO methods (Hashimoto et al., 2018; Sagawa et al., 2020) primarily address performance disparities arising from natural shifts across demographic groups. Subsequent work extends this framework to min-max group fairness (Diana et al., 2021) and scalable DRO formulations (Levy et al., 2020). Wang et al. (2023) develop the CUMA algorithm to maintain fairness under unseen domains by matching loss curvatures across demographic groups. These approaches provide algorithmic guarantees under clean-data assumptions, but do not address arbitrary data corruption.

Another line of work considers robustness to data poisoning attacks. Roh et al. (2020) propose FR-Train, a mutual information-based approach that maintains group fairness while defending against data poisoning by identifying and reducing the influence of malicious samples. Wang et al. (2020) address noisy protected group labels via robust optimization, providing guarantees for true group fairness under statistical noise; however, they assume a probabilistic noise model rather than adversarial corruption and do not address individual fairness. Neither approach provides corruption-calibrated theoretical guarantees under the α -corruption model.

In contrast to the above, the α -corruption model (Valiant, 1985; Kearns & Li, 1993) allows training samples to be arbitrarily corrupted within a bounded fraction. Under this model, Konstantinov & Lampert (2022) establish hardness results showing that any polynomial-time learner achieving small excess risk must incur demographic parity violations scaling as $\Omega(\alpha/\pi_0)$, inversely proportional to the minority group size. Blum et al. (2024) analyze randomized classifiers under label-flipping attacks, demonstrating $\Theta(\alpha)$ accuracy degradation while approximately preserving demographic parity, at the cost of requiring access to sensitive attributes at test time. While these results provide tight impossibility and vulnerability bounds for group fairness under corruption, they do not yield constructive algorithms or address individual fairness.

Recent work explores alternative robustness paradigms for fair learning. Causal DRO approaches (Ehyaei et al., 2024) use Wasserstein-based uncertainty sets with causal trans-

port metrics to enforce counterfactual individual fairness, addressing a complementary setting where causal structure is known. Xu et al. (2022) propose a framework for Robust Fair Classification (RFC) as a defense against data poisoning attacks targeting fairness, using subgroup sanitization to identify and remove malicious samples. Our work differs in using TV-based ambiguity sets calibrated directly from the corruption rate α , providing explicit robustness guarantees without requiring causal graphs or sanitization procedures.

Table 6 (Appendix B) summarizes the positioning of our work relative to prior approaches.

3. Preliminaries

In this section, we describe the setup and introduce relevant notation, definitions, and theorems.

3.1. Fair Classification Setup

We consider fair binary classification on the domain $\mathcal{Z} = \mathcal{X} \times \mathcal{A} \times \mathcal{Y}$, where $\mathcal{X}$ denotes the feature space, $\mathcal{A} = \{0, 1\}$ the protected attribute space, and $\mathcal{Y} = \{0, 1\}$ the label space. Let (X, A, Y) denote random variables over $\mathcal{Z}$ with realizations (x, a, y) . We assume data is drawn i.i.d. from a distribution $\mathbb{P}$ over $\mathcal{Z}$, with marginal $\mathbb{P}_X$ on $\mathcal{X}$. Let $\pi_j = \mathbb{P}(A = j)$ denote the proportion of group $j \in \{0, 1\}$; we assume $\pi_0, \pi_1 > 0$ (both groups exist in the population) and, without loss of generality, that $\pi_0 \leq \pi_1$, i.e., group 0 is the minority group. The learner selects a classifier $h : \mathcal{X} \rightarrow \{0, 1\}$ from a hypothesis class $\mathcal{H} = \{h_\theta : \theta \in \Theta\}$ with finite VC dimension D_{VC} , where Θ denotes the parameter space and h_θ is the classifier corresponding to parameter θ .

We focus on two well-established fairness criteria: *Demographic Parity*, ensuring equal positive classification rates across groups, and *Individual Fairness*, requiring similar individuals to receive similar predictions.

Definition 3.1 (Demographic Parity (DP) (Calders et al., 2009)). A classifier h satisfies DP if $p_0 = p_1$, where $p_a = \mathbb{P}(h(X) = 1 \mid A = a)$ for $a \in \{0, 1\}$.

The DP violation (Calders & Verwer, 2010) is:

$$\Delta_{DP}(h; \mathbb{P}) := |p_0 - p_1|. \quad (1)$$

Definition 3.2 (Individual Fairness (IF) (Dwork et al., 2012)). Let $d : \mathcal{X} \times \mathcal{X} \rightarrow [0, 1]$ be a domain-specific distance metric. A classifier h satisfies (d, γ) -metric fairness with slack $\gamma \geq 0$ if

$$|h(x) - h(x')| \leq d(x, x') + \gamma \quad \forall x, x' \in \mathcal{X}.$$

The IF violation (Rothblum & Yona (2018)) is:

$$L_{d, \gamma}(h; \mathbb{P}) := \mathbb{E}_{X, X' \sim \mathbb{P}_X} \left[\mathbb{1}(|h(X) - h(X')| > d(X, X') + \gamma) \right], \quad (2)$$

where X and X' are independent samples from $\mathbb{P}_X$.

Classification and Empirical Objectives. Let $R(h; \mathbb{P}) = \mathbb{P}(h(X) \neq Y)$ denote the classification error under the 0-1 loss. Given an i.i.d. sample $S = \{(x_i, a_i, y_i)\}_{i=1}^n$ from $\mathbb{P}$, the empirical classification error is:

$$\hat{R}(h; S) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}(h(x_i) \neq y_i).$$

Similarly, let $\hat{\Delta}_{DP}(h; S)$ and $\hat{L}_{d, \gamma}(h; S)$ denote the empirical Demographic Parity and Individual fairness violation computed on sample S (see Appendix A for explicit formulas).

Uniform Convergence Guarantees. Our generalization bounds are based on the classical VC-based uniform convergence theorem (Vapnik & Chervonenkis, 1971; Blumer et al., 1989).

Theorem 3.3. (Shalev-Shwartz & Ben-David (2014)) Let $\mathcal{H}$ have VC dimension D_{VC} . For any $\delta > 0$, with probability at least $1 - \delta$ over an i.i.d. sample S of size n from $\mathbb{P}$,

$$\sup_{h \in \mathcal{H}} |R(h; \mathbb{P}) - \hat{R}(h; S)| \leq 2 \sqrt{\frac{8D_{VC} \log(en/D_{VC}) + 2 \log(4/\delta)}{n}}.$$

This theorem provides the foundation for establishing generalization guarantees for our formulations.

4. Fairness under the α -Corruption Model

Real world training data often contains corruptions in features, protected attributes, or labels arising from measurement errors, data collection biases, or other sources. We formalize this using the α -corruption model (Valiant, 1985; Kearns & Li, 1993; Cesa-Bianchi et al., 1999).

Definition 4.1 (α -Corruption Model). Let $\mathbb{P}$ denote the clean distribution over $\mathcal{Z}$, and let $S = \{(x_i, a_i, y_i)\}_{i=1}^n$ be an i.i.d. sample from $\mathbb{P}$. For corruption rate $\alpha \in [0, 1/2)$, each index $i \in [n]$ is independently selected with probability α , forming the corruption set $M \subseteq [n]$. For each $i \in M$, the sample (x_i, a_i, y_i) is replaced with (x'_i, a'_i, y'_i) drawn from an arbitrary distribution $\mathbb{Q}$. The learner observes the corrupted sample $S^c = \{(x_i^c, a_i^c, y_i^c)\}_{i=1}^n$, inducing the mixture distribution:

$$\mathbb{P}_{\text{pert}} = (1 - \alpha) \mathbb{P} + \alpha \mathbb{Q}. \quad (3)$$

The constraint $\alpha < 1/2$ ensures that clean samples remain the majority, preserving learnability.

We quantify corruption's impact via total variation bounds between clean and perturbed distributions for group-conditional distributions (Theorem 4.2) and feature product measures (Theorem 4.3).

Theorem 4.2 (Group-Conditional TV Bound). *Under the α -corruption model (Definition 4.1), for each group $j \in \{0, 1\}$ with $\pi_j > 0$ (as assumed in Section 3.1), the total variation distance between the clean and perturbed group-conditional distributions satisfies:*

$$\text{TV}(\mathbb{P}_{X,Y|A=j}, \mathbb{P}_{\text{pert},X,Y|A=j}) \leq \frac{\alpha}{(1-\alpha)\pi_j + \alpha}. \quad (4)$$

Proof sketch. Express the perturbed conditional as a mixture $\mathbb{P}_{\text{pert},j} = w\mathbb{P}_j + (1-w)\mathbb{Q}_j$, apply TV convexity to obtain $\text{TV}(\mathbb{P}_j, \mathbb{P}_{\text{pert},j}) = 1-w$, then maximize over corruption group proportions $q_j \in [0, 1]$. (Appendix C.1.) $\square$

Theorem 4.3 (Product Measure TV Bound). *Under the α -corruption model (Definition 4.1), the total variation distance between the product measures of the clean and perturbed feature marginals satisfies:*

$$\text{TV}(\mathbb{P}_X^{\otimes 2}, \mathbb{P}_{\text{pert},X}^{\otimes 2}) \leq 2\alpha - \alpha^2. \quad (5)$$

Proof sketch. Construct a coupling between $(X_1, X_2) \sim \mathbb{P}_X^{\otimes 2}$ and $(Z_1, Z_2) \sim \mathbb{P}_{\text{pert},X}^{\otimes 2}$ where each Z_i is corrupted independently with probability α , yielding disagreement probability $1 - (1-\alpha)^2 = 2\alpha - \alpha^2$. (Appendix C.2.) $\square$

The total-variation bounds in Theorems 4.2 and 4.3 enable us to bound how much group-conditional rates and individual fairness violation can change under corruption.

Corollary 4.4 (Group-Conditional Rate Stability). *Under the α -corruption model (Definition 4.1), for any classifier $h : \mathcal{X} \rightarrow \{0, 1\}$ and each group $j \in \{0, 1\}$ with $\pi_j > 0$, the group-conditional positive classification rate satisfies:*

$$|\mathbb{P}(h(X) = 1 \mid A = j) - \mathbb{P}_{\text{pert}}(h(X) = 1 \mid A = j)| \leq \frac{\alpha}{(1-\alpha)\pi_j + \alpha}. \quad (6)$$

Proof sketch. Apply Theorem 4.2 to the event $B = \{(x, y) : h(x) = 1\}$, noting that the shift in acceptance rate $|\mathbb{P}_j(B) - \mathbb{P}_{\text{pert},j}(B)|$ is bounded by the TV distance. (Appendix C.3.) $\square$

Corollary 4.5 (Individual Fairness Stability). *Under the α -corruption model (Definition 4.1), for any binary classifier $h : \mathcal{X} \rightarrow \{0, 1\}$, the difference between metric fairness violations under clean and perturbed distributions satisfies:*

$$|L_{d,\gamma}(h; \mathbb{P}) - L_{d,\gamma}(h; \mathbb{P}_{\text{pert}})| \leq 2\alpha - \alpha^2. \quad (7)$$

Proof sketch. Express the IF violation as $L_{d,\gamma}(h; \mathbb{P}) = \mathbb{E}_{\mathbb{P}_{\otimes^2}^X}[\phi]$ for indicator $\phi \in [0, 1]$, then bound expectation difference by TV distance via Theorem 4.3. (Appendix C.4.) $\square$

5. Naive-FAIR: Naive Approach

We first analyze a baseline approach that treats the corrupted training data as if it were clean, establishing bounds on how fairness degrades without robustness measures.

5.1. Naive-FAIR Formulation

For the parameterized classifier h_θ trained on S^c , the constrained optimization problem is:

$$\begin{aligned} \min_{\theta} \quad & \hat{R}(h_\theta; S^c) \\ \text{s.t.} \quad & \hat{\Delta}_{\text{DP}}(h_\theta; S^c) \leq \varepsilon_g, \\ & \hat{L}_{d,\gamma}(h_\theta; S^c) \leq \varepsilon_i, \end{aligned} \quad (8)$$

where $\varepsilon_g, \varepsilon_i > 0$ are tolerance parameters for group and individual fairness, respectively.

Introducing Lagrange multipliers $\lambda_{\text{DP}}, \lambda_{\text{IF}} \geq 0$, we obtain the objective for Naive-FAIR:

$$\begin{aligned} \hat{\mathcal{L}}_{\text{naive}}(h_\theta, \lambda_{\text{DP}}, \lambda_{\text{IF}}; S^c) = & \hat{R}(h_\theta; S^c) + \lambda_{\text{DP}} \hat{\Delta}_{\text{DP}}(h_\theta; S^c) \\ & + \lambda_{\text{IF}} \hat{L}_{d,\gamma}(h_\theta; S^c). \end{aligned} \quad (9)$$

By standard Lagrangian duality, Naive-FAIR solves the following min-max optimization:

$$\min_{\theta} \max_{\lambda_{\text{DP}}, \lambda_{\text{IF}} \geq 0} \hat{\mathcal{L}}_{\text{naive}}(h_\theta, \lambda_{\text{DP}}, \lambda_{\text{IF}}; S^c). \quad (10)$$

5.2. Sample Complexity of Naive-FAIR Lagrangian Objective

We now characterize the generalization properties of the Naive-FAIR Lagrangian. The corresponding population-level objective is:

$$\begin{aligned} \mathcal{L}_{\text{naive}}(h, \lambda_{\text{DP}}, \lambda_{\text{IF}}; \mathbb{P}) = & R(h; \mathbb{P}) + \lambda_{\text{DP}} \Delta_{\text{DP}}(h; \mathbb{P}) \\ & + \lambda_{\text{IF}} L_{d,\gamma}(h; \mathbb{P}). \end{aligned} \quad (11)$$

Theorem 5.1 (Sample Complexity of Naive-FAIR Lagrangian). *Let $\mathcal{H}$ have VC dimension D_{VC} , and let S^c be an α -corrupted sample of size n from $\mathbb{P}_{\text{pert}}$ (Definition 4.1). Let π_0 denote the minority group proportion. Let $\hat{h}$ denote the empirical minimizer:*

$$\hat{h} = \arg \min_{h \in \mathcal{H}} \hat{\mathcal{L}}_{\text{naive}}(h; S^c).$$

For any $\delta \in (0, 1)$, if

$$n \geq \max \left\{ \frac{10 \log(6/\delta)}{\alpha}, \frac{8 \log(12/\delta)}{(1-\alpha)\pi_0}, \frac{2D_{\text{VC}}}{(1-\alpha)\pi_0} \right\},$$

then with probability at least $1 - \delta$,

$$\mathcal{L}_{\text{naive}}(\hat{h}; \mathbb{P}) \leq \min_{h \in \mathcal{H}} \mathcal{L}_{\text{naive}}(h; \mathbb{P}) + \Delta_\lambda, \quad (12)$$

where the excess risk is

$$\begin{aligned} \Delta_\lambda \leq & 3\alpha + 2\lambda_{\text{IF}}(2\alpha - \alpha^2) + \frac{4\lambda_{\text{DP}}\alpha}{(1 - \alpha)\pi_0 + \alpha} \\ & + (1 + \lambda_{\text{IF}}) \cdot O\left(\sqrt{\frac{D_{\text{VC}} \log n + \log(1/\delta)}{n}}\right) \\ & + \lambda_{\text{DP}} \cdot O\left(\sqrt{\frac{D_{\text{VC}} \log n + \log(1/\delta)}{(1 - \alpha)\pi_0 n}}\right). \end{aligned} \quad (13)$$

Proof sketch. We decompose the Lagrangian into three components (accuracy, DP, IF), bound each uniformly via VC theory under corruption, then combine via union bound and ERM symmetrization. See Appendix D. $\square$

6. DRO-FAIR: Robust Approach

While Theorem 5.1 bounds quantify how fairness degrades under corruption, they do not ensure fairness on the clean distribution $\mathbb{P}$. We therefore adopt distributionally robust optimization (DRO) (Shapiro et al., 2014; Rahimian & Mehrotra, 2022) to provide robust fairness guarantees.

6.1. Corruption-Calibrated Uncertainty Sets

To provide guarantees for the clean distribution $\mathbb{P}$, we construct uncertainty sets around the corrupted distribution. These sets are calibrated using our total-variation bounds (Theorems 4.2 and 4.3) to ensure $\mathbb{P}$ lies within them by design.

Let $\hat{p}_n \in \Delta^n$ denote the uniform distribution over the corrupted sample $S^c = \{(x_i^c, a_i^c, y_i^c)\}_{i=1}^n$.¹ For each group $j \in \{0, 1\}$, let $\hat{p}_{n_j} \in \Delta^{n_j}$ denote the uniform distribution over the subset of S^c where $a_i^c = j$. To capture distributional shifts, we introduce reweighting vectors: $(\tilde{p}_0, \tilde{p}_1) \in \Delta^{n_0} \times \Delta^{n_1}$ for group-conditional shifts and $\tilde{p} \in \Delta^n$ for global shifts.

Using these reweighting vectors, we define the weighted empirical fairness violations as:

$$\begin{aligned} \hat{\Delta}_{\text{DP}}(h; S^c, \tilde{p}_0, \tilde{p}_1) := & \left| \sum_{i: a_i^c=1} \tilde{p}_{1,i} h(x_i^c) - \sum_{i: a_i^c=0} \tilde{p}_{0,i} h(x_i^c) \right|. \end{aligned} \quad (14)$$

¹ $\Delta^k = \{p \in \mathbb{R}^k : p_i \geq 0, \sum_i p_i = 1\}$ denotes the probability simplex.

$$\begin{aligned} \hat{L}_{d,\gamma}(h; S^c, \tilde{p}) := & \frac{1}{n-1} \sum_{i \neq j} \frac{\tilde{p}_i + \tilde{p}_j}{2} \\ & \cdot \mathbb{1}[|h(x_i^c) - h(x_j^c)| > d(x_i^c, x_j^c) + \gamma]. \end{aligned} \quad (15)$$

The factored weight $(\tilde{p}_i + \tilde{p}_j)/2$ is justified in Appendix E.1.

When $\tilde{p}_j = \hat{p}_{n_j}$ (uniform over group j) for both $j \in \{0, 1\}$, $\hat{\Delta}_{\text{DP}}(h; S^c, \hat{p}_0, \hat{p}_1) = \hat{\Delta}_{\text{DP}}(h; S^c)$. Similarly, $\hat{L}_{d,\gamma}(h; S^c, \hat{p}_n) = \hat{L}_{d,\gamma}(h; S^c)$.

We define uncertainty sets as ℓ_1 -balls around the empirical distributions, leveraging the identity $\|P - Q\|_1 = 2 \text{TV}(P, Q)$. The radii are set using the total-variation bounds from Theorems 4.2 and 4.3:

$$\begin{aligned} \mathcal{U}_{\text{DP}} &= \mathcal{U}_0 \times \mathcal{U}_1, \\ \mathcal{U}_j &= \{\tilde{p} \in \Delta^{n_j} : \|\tilde{p} - \hat{p}_{n_j}\|_1 \leq 2\rho_{\text{DP},j}\}, \\ \mathcal{U}_{\text{IF}} &= \{\tilde{p} \in \Delta^n : \|\tilde{p} - \hat{p}_n\|_1 \leq 2\rho_{\text{IF}}\}, \\ \rho_{\text{DP},j} &= \frac{\alpha}{(1 - \alpha)\pi_j + \alpha}, \quad \rho_{\text{IF}} = 2\alpha - \alpha^2. \end{aligned} \quad (16)$$

Crucially, the group-specific radii $\rho_{\text{DP},j}$ scale inversely with group proportion π_j , reflecting the greater vulnerability of minority groups under corruption.

6.2. DRO Formulation

We now formalize the robust optimization problem. For the parameterized classifier h_θ trained on the corrupted sample S^c , the constrained DRO formulation is:

$$\begin{aligned} \min_{\theta} \quad & \hat{R}(h_\theta; S^c) \\ \text{s.t.} \quad & \max_{(\tilde{p}_0, \tilde{p}_1) \in \mathcal{U}_{\text{DP}}} \hat{\Delta}_{\text{DP}}(h_\theta; S^c, \tilde{p}_0, \tilde{p}_1) \leq \varepsilon_g, \\ & \max_{\tilde{p} \in \mathcal{U}_{\text{IF}}} \hat{L}_{d,\gamma}(h_\theta; S^c, \tilde{p}) \leq \varepsilon_i, \end{aligned} \quad (17)$$

where $\varepsilon_g, \varepsilon_i > 0$ are tolerance parameters for group and individual fairness, respectively. Unlike the naive formulation (8), these constraints must hold under worst-case reweighting within the uncertainty sets.

Introducing Lagrange multipliers $\lambda_{\text{DP}}, \lambda_{\text{IF}} \geq 0$, we obtain the robust fairness Lagrangian:

$$\begin{aligned} \hat{\mathcal{L}}_{\text{DRO}}(h_\theta, \lambda_{\text{DP}}, \lambda_{\text{IF}}; S^c, \mathcal{U}_{\text{DP}}, \mathcal{U}_{\text{IF}}) &= \hat{R}(h_\theta; S^c) \\ &+ \lambda_{\text{DP}} \max_{(\tilde{p}_0, \tilde{p}_1) \in \mathcal{U}_{\text{DP}}} \hat{\Delta}_{\text{DP}}(h_\theta; S^c, \tilde{p}_0, \tilde{p}_1) \\ &+ \lambda_{\text{IF}} \max_{\tilde{p} \in \mathcal{U}_{\text{IF}}} \hat{L}_{d,\gamma}(h_\theta; S^c, \tilde{p}). \end{aligned} \quad (18)$$

By standard Lagrangian duality, the constrained problem (17) is equivalent to a min-max saddle-point formulation. DRO-FAIR solves the following optimization:

$$\min_{\theta} \max_{\lambda_{\text{DP}}, \lambda_{\text{IF}} \geq 0} \hat{\mathcal{L}}_{\text{DRO}}(h_\theta, \lambda_{\text{DP}}, \lambda_{\text{IF}}; S^c, \mathcal{U}), \quad (19)$$

where we denote the combined uncertainty sets as $\mathcal{U} = (\mathcal{U}_{\text{DP}}, \mathcal{U}_{\text{IF}})$ for compactness.

6.3. Sample Complexity of DRO Lagrangian Objective

We now characterize the generalization properties of the DRO-FAIR Lagrangian. Recall the empirical DRO objective (18). The corresponding population-level uncertainty sets are defined as

$$\begin{aligned}\mathcal{B}_{\text{DP},j} &= \{Q : \text{TV}(Q, \mathbb{P}_{\text{pert},X,Y|A=j}) \leq \rho_{\text{DP},j}\}, \\ j &\in \{0, 1\}, \\ \mathcal{B}_{\text{IF}} &= \{Q : \text{TV}(Q, \mathbb{P}_{\text{pert},X}^{\otimes 2}) \leq \rho_{\text{IF}}\}.\end{aligned}$$

with $\mathcal{B}_{\text{DP}} = \mathcal{B}_{\text{DP},0} \times \mathcal{B}_{\text{DP},1}$ and $\mathcal{B} = \mathcal{B}_{\text{DP}} \times \mathcal{B}_{\text{IF}}$. The population DRO objective is:

$$\mathcal{L}_{\text{DRO}}(h, \lambda_{\text{DP}}, \lambda_{\text{IF}}; \mathbb{P}, \mathcal{B}) = R(h; \mathbb{P}) + \lambda_{\text{DP}} \sup_{Q \in \mathcal{B}_{\text{DP}}} \Delta_{\text{DP}}(h; Q) + \lambda_{\text{IF}} \sup_{Q \in \mathcal{B}_{\text{IF}}} L_{d,\gamma}(h; Q). \quad (20)$$

Here $R(h; \mathbb{P})$ is the classification risk on the clean distribution, while $\mathcal{B}$ is centered at the corrupted distribution $\mathbb{P}_{\text{pert}}$. By our radius calibration (Theorems 4.2 and 4.3), we have $\mathbb{P} \in \mathcal{B}$, ensuring the worst-case fairness terms provide upper bounds on violations at $\mathbb{P}$.

Theorem 6.1 (Sample Complexity of DRO-FAIR). *Let $\mathcal{H}$ have VC dimension D_{VC} , and let S^c be an α -corrupted sample of size n from $\mathbb{P}_{\text{pert}}$ (Definition 4.1). Let π_0 denote the minority group proportion. Let $\hat{h}$ denote the empirical minimizer:*

$$\hat{h} = \arg \min_{h \in \mathcal{H}} \hat{\mathcal{L}}_{\text{DRO}}(h; S^c, \mathcal{U}).$$

For any $\delta \in (0, 1)$, if

$$n \geq \max \left\{ \frac{10 \log(6/\delta)}{\alpha}, \frac{8 \log(12/\delta)}{(1-\alpha)\pi_0}, \frac{2D_{\text{VC}}}{(1-\alpha)\pi_0} \right\}, \quad (21)$$

then with probability at least $1 - \delta$,

$$\mathcal{L}_{\text{DRO}}(\hat{h}; \mathbb{P}, \mathcal{B}) \leq \min_{h \in \mathcal{H}} \mathcal{L}_{\text{DRO}}(h; \mathbb{P}, \mathcal{B}) + \Delta_{\text{DRO}}, \quad (22)$$

where the excess risk is

$$\begin{aligned}\Delta_{\text{DRO}} &\leq 3\alpha + 2\lambda_{\text{IF}}(2\alpha - \alpha^2) + \frac{4\lambda_{\text{DP}}\alpha}{(1-\alpha)\pi_0 + \alpha} \\ &\quad + (1 + \lambda_{\text{IF}}) \cdot O\left(\sqrt{\frac{D_{\text{VC}} \log n + \log(1/\delta)}{n}}\right) \\ &\quad + \lambda_{\text{DP}} \cdot O\left(\sqrt{\frac{D_{\text{VC}} \log n + \log(1/\delta)}{(1-\alpha)\pi_0 n}}\right).\end{aligned} \quad (23)$$

Proof sketch. We characterize worst-case fairness violations by introducing an approximation function F that achieves radius-bounded error for both empirical and population worst-case DP (handling edge cases where classifiers are constant on samples but non-constant on the population). For IF, we reduce the pairwise weighted objective to a linear form via symmetry, enabling TV-ball robust expectation bounds. Standard VC uniform convergence (with U-process concentration for the pairwise IF term) bounds the empirical-to-population gap, and ERM symmetrization yields the final excess risk bound. (Appendix E.1.) $\square$

6.4. Interpretation and Comparison

The generalization bounds for Naive-FAIR (Theorem 5.1) and DRO-FAIR (Theorem 6.1) decompose as: (Empirical violation) + (Corruption penalty) + (Statistical error). The key distinction lies in the empirical term: **Naive-FAIR** uses $\hat{\Delta}_{\text{DP}}$ and $\hat{L}_{d,\gamma}$ computed directly on S^c , which may be optimistically small due to favorable corruption realizations; **DRO-FAIR** uses the worst-case $\max_{(\tilde{p}_0, \tilde{p}_1) \in \mathcal{U}_{\text{DP}}} \hat{\Delta}_{\text{DP}}$ and $\max_{\tilde{p} \in \mathcal{U}_{\text{IF}}} \hat{L}_{d,\gamma}$, guarding against adversarial corruptions. The corruption penalty terms $O(\lambda_{\text{DP}}\rho_{\text{DP},0})$ and $O(\lambda_{\text{IF}}\rho_{\text{IF}})$ in the excess risk bounds are unavoidable: they capture the information-theoretic cost of ensuring fairness on the clean distribution $\mathbb{P}$ when only the corrupted distribution $\mathbb{P}_{\text{pert}}$ is observed. These penalties arise because (i) the clean distribution lies within TV-distance ρ from the corrupted distribution (Theorems 4.2–4.3), and (ii) when a classifier is empirically constant on the sample but non-constant on the population, reweighting cannot detect this discrepancy, requiring an additive correction of order ρ (Lemma E.5). **Trade-offs:** DRO-FAIR provides robust guarantees but requires min-max optimization and knowledge of α ; when α is unknown, Naive-FAIR provides baseline guarantees.

Remark 6.2 (Radius Optimality). The radii $\rho_{\text{DP},j} = \frac{\alpha}{(1-\alpha)\pi_j + \alpha}$ and $\rho_{\text{IF}} = 2\alpha - \alpha^2$ are tight: (i) *Necessity*: smaller radii would exclude $\mathbb{P}$ from $\mathcal{B}$ for some adversarial $\mathbb{Q}$, invalidating the robustness guarantee; (ii) *Achievability*: bounds are attained when $\mathbb{Q}$ concentrates on the opposite group (DP) or disjoint support (IF). Using radii larger than necessary makes the worst-case constraints more conservative without improving the guarantee (see Appendix J for empirical validation).

7. Experiments and Results

In this section, to empirically validate the theoretical properties of DRO-FAIR framework we answer the following questions:

- **RQ1 (Robustness):** Does DRO-FAIR maintain fairness constraints (both DP and IF) more effectively than

Table 1. **Quantitative Results.** Mean $\pm$ SE over 10 seeds. Bold = best per metric. DRO-FAIR achieves substantial fairness improvements with moderate accuracy cost. Training: Naive-FAIR $\sim$ 5s, DRO-FAIR $\sim$ 60s per run (A100 GPU).

α	Method	Acc $\uparrow$	Adult DP $\downarrow$	IF $\downarrow$	Acc $\uparrow$	Credit DP $\downarrow$	IF $\downarrow$	Acc $\uparrow$	LSAC DP $\downarrow$	IF $\downarrow$
0.0	Naive	.814 $\pm$.002	.008 $\pm$.003	.043 $\pm$.001	.817 $\pm$.001	.020 $\pm$.003	.035 $\pm$.002	.907 $\pm$.001	.006 $\pm$.001	.012 $\pm$.002
	DRO-FAIR	.813 $\pm$.001	.005 $\pm$.001	.043 $\pm$.002	.817 $\pm$.001	.017 $\pm$.002	.034 $\pm$.001	.908 $\pm$.001	.008 $\pm$.002	.014 $\pm$.002
0.1	Naive	.835 $\pm$.001	.107 $\pm$.005	.064 $\pm$.001	.815 $\pm$.002	.019 $\pm$.002	.037 $\pm$.001	.906 $\pm$.002	.010 $\pm$.002	.016 $\pm$.002
	DRO-FAIR	.815 $\pm$.005	.058 $\pm$.020	.058 $\pm$.005	.791 $\pm$.005	.010 $\pm$.003	.013 $\pm$.004	.901 $\pm$.001	.007 $\pm$.002	.007 $\pm$.002
0.2	Naive	.833 $\pm$.001	.168 $\pm$.005	.067 $\pm$.002	.815 $\pm$.001	.020 $\pm$.002	.036 $\pm$.002	.901 $\pm$.001	.010 $\pm$.003	.015 $\pm$.004
	DRO-FAIR	.795 $\pm$.002	.028 $\pm$.007	.045 $\pm$.002	.782 $\pm$.002	.003 $\pm$.001	.007 $\pm$.003	.897 $\pm$.003	.014 $\pm$.007	.005 $\pm$.001
0.3	Naive	.828 $\pm$.001	.229 $\pm$.010	.077 $\pm$.001	.813 $\pm$.002	.024 $\pm$.003	.035 $\pm$.001	.900 $\pm$.002	.006 $\pm$.001	.018 $\pm$.002
	DRO-FAIR	.786 $\pm$.003	.065 $\pm$.036	.052 $\pm$.004	.788 $\pm$.004	.009 $\pm$.003	.012 $\pm$.004	.891 $\pm$.003	.026 $\pm$.007	.006 $\pm$.001
0.4	Naive	.816 $\pm$.002	.290 $\pm$.008	.078 $\pm$.001	.813 $\pm$.001	.027 $\pm$.002	.034 $\pm$.001	.896 $\pm$.002	.009 $\pm$.002	.017 $\pm$.002
	DRO-FAIR	.753 $\pm$.013	.333 $\pm$.057	.065 $\pm$.005	.802 $\pm$.003	.025 $\pm$.004	.025 $\pm$.004	.888 $\pm$.002	.029 $\pm$.006	.011 $\pm$.003

Naive-FAIR as the corruption rate α increases?

- **RQ2 (Trade-offs):** What is the accuracy cost for achieving robust fairness?
- **RQ3 (Efficiency):** What is the computational overhead of DRO-FAIR compared to Naive-FAIR?

We further analyze the importance weights learnt in DRO-FAIR and provide ablation results.

7.1. Experimental Setup

Datasets. We evaluate on three standard fairness benchmarks: **Adult** (Becker & Kohavi, 1996) (45,222 samples, 12 features, predicting income $>$ \$50K with *sex* as protected attribute), **Credit Card Default** (Yeh, 2009) (30,000 samples, 22 features, predicting payment default with *sex* as protected attribute), and **LSAC Law School** (Wightman, 1998) (18,692 samples, 10 features, predicting bar passage with *sex* as protected attribute). Further details are provided in Appendix G.5.

Corruption Protocol. To simulate the α -corruption model (Section 4), we apply comprehensive corruption to the training and validation sets with rate $\alpha \in \{0.0, 0.1, 0.2, 0.3, 0.4\}$. For each selected sample (with probability α), we add Gaussian noise $\mathcal{N}(0, 0.1\sigma_{\text{col}})$ to numeric features (where σ_{col} is the column-wise standard deviation), replace categorical features with a uniformly random different category, flip labels ($0 \leftrightarrow 1$), and flip the protected attribute to a different group value. Test sets remain uncorrupted to evaluate clean distribution performance. Our theoretical bounds hold for worst-case adversarial corruption, while experiments use random corruption as a realistic lower bound (see Appendix F).

Baseline and Optimization. We compare our proposed method against Naive-FAIR (Section 5). This baseline naively assumes that the training data contains no corruption. We use AdamW (Loshchilov & Hutter, 2019) for θ -updates

with gradient clipping, Adam for $\tilde{p}$ -updates, and clamp $\lambda \in [0, \lambda_{\max}]$ for stability. We set $K = 10$ inner steps and $k = 5$ for the k -NN graph. We use $\tau = 100$ and $\beta = 5$ for all datasets and corruption rates. All experiments are conducted over 10 random seeds to ensure statistical reliability; we report mean $\pm$ standard error (SE).

7.2. Results

Quantitative results are summarized in Table 1.

RQ1: Robustness to Corruption. Across all datasets and moderate corruption levels ($\alpha \leq 0.3$), DRO-FAIR achieves substantially lower fairness violations than Naive-FAIR. On Adult, DRO-FAIR reduces DP violation by up to 83% at $\alpha = 0.2$ ($0.168 \rightarrow 0.028$) and maintains consistent IF improvements (33% lower at $\alpha = 0.2$). On Credit (Table 1) DRO-FAIR achieves $5\times$ lower IF violations at $\alpha = 0.2$. On LSAC, DRO-FAIR shows strong IF improvements but exhibits DP-IF trade-offs at $\alpha \geq 0.3$: $3\times$ lower IF but $4\times$ higher DP, reflecting the inherent tension between group and individual fairness constraints (Table 4). At high corruption ($\alpha = 0.4$), both methods exhibit increased variance; tuning $\tau = 1$ on Adult recovers strong performance (95% lower DP, 68% lower IF versus Naive-FAIR).

RQ2: Accuracy-Fairness Trade-off. DRO-FAIR incurs a moderate accuracy cost (1-4%) to achieve fairness. This trade-off is favorable: at $\alpha = 0.3$ on Adult, a 4% accuracy reduction yields 72% lower DP violation ($0.229 \rightarrow 0.065$). On LSAC, DRO-FAIR achieves $2\times$ lower IF violations at $\alpha = 0.1$ with only 0.5% accuracy cost.

RQ3: Computational Overhead. DRO-FAIR requires approximately $12\times$ longer training time than Naive-FAIR (~ 60 s vs ~ 5 s per run on A100 GPU) due to the inner maximization over uncertainty sets. This overhead is acceptable for offline training where robust fairness is critical.

Table 2. Similarity between learned IF and DP weights by subgroup.

Subgroup	Spearman R	Cosine Sim.
Global (All Data)	0.0127	0.1606
Female (Unpriv.), Pos	0.0157	0.0539
Male (Priv.), Neg	0.2064	0.2196
Male (Priv.), Pos	0.1206	0.1759
Female (Unpriv.), Neg	0.0023	0.0779

Table 3. Mean learned importance weights by subgroup. The last column shows the mean per-sample absolute difference $\mathbb{E}[\|\bar{w}_i^{\text{DP}} - \bar{w}_i^{\text{IF}}\|]$.

Subgroup	IF Weight	DP Weight	Mean $ \Delta\bar{w} $
Female, Pos	1.25 ± 0.20	0.76 ± 0.07	1.43
Male, Neg	0.90 ± 0.06	0.86 ± 0.03	0.46
Male, Pos	1.27 ± 0.14	1.27 ± 0.07	1.19
Female, Neg	0.86 ± 0.07	1.07 ± 0.04	0.92

7.3. Analysis of Learned Weights

To better understand how DRO-FAIR achieves robustness, we analyze the importance weights learned during optimization on the Credit dataset with $\alpha = 0.2$. The re-weighting mechanism upweights “hard” samples to identify the worst-case distribution within the uncertainty set. For interpretability, we report **normalized importance weights** centered at 1.0: for IF, we normalize globally as $\bar{w}_i^{\text{IF}} = n \cdot \tilde{p}_i$; for DP, we normalize within each group as $\bar{w}_i^{\text{DP}} = n_j \cdot \tilde{p}_{j,i}$ for sample i in group j . A weight $\bar{w}_i > 1$ indicates a sample is upweighted relative to the uniform distribution.

Weight Similarity. We first examine whether the optimization identifies similar samples as “hard” for both Individual Fairness (IF) and Demographic Parity (DP). Table 2 presents the Spearman correlation and Cosine similarity between the learned IF and DP weights. The low global correlation (0.0127) suggests that the two fairness notions prioritize different subsets of the data, highlighting the importance of the joint optimization in Eq. (19).

Subgroup Analysis. We further aggregate the weights by subgroup (Table 3). For IF, the optimization assigns the highest weights to *Male, Pos* (1.27) and *Female, Pos* (1.25). For DP, the weights focus on *Male, Pos* (1.27) and *Female, Neg* (1.07). The disparity in weight distribution is most pronounced for the *Female, Pos* subgroup, with a mean per-sample absolute difference of 1.43.

7.4. Ablation Studies

DP-IF Constraint Ablation. To investigate the DP-IF trade-off observed in Table 1, we conduct an ablation study removing the IF constraint from both DRO-FAIR

Table 4. Ablation: Effect of removing IF constraint on LSAC ($\alpha = 0.4$).

Method	Acc $\uparrow$	DP $\downarrow$	IF $\downarrow$
Naive-FAIR (with IF)	.896	.009	.017
Naive-FAIR (no IF)	.897	.009	.025
DRO-FAIR (with IF)	.888	.029	.011
DRO-FAIR (no IF)	.897	.020	.027

Table 5. Ablation: Effect of radius selection on Credit dataset.

α	Radii	Acc	DP $\downarrow$	IF $\downarrow$
0.1	Calibrated	.791	.010	.013
	$\rho_{\text{DP},j} = \rho_{\text{IF}} = 1$	.789	.016	.040
0.2	Calibrated	.782	.003	.007
	$\rho_{\text{DP},j} = \rho_{\text{IF}} = 1$	.808	.025	.025

and Naive-FAIR on LSAC at $\alpha = 0.4$ (Table 4). Removing the IF constraint improves DRO-FAIR’s DP from 0.029 to 0.020 and accuracy from 0.888 to 0.897, confirming that the IF constraint competes with DP optimization. This explains the DP-IF trade-off observed in the main results for LSAC at higher corruption levels. When constraints compete, practitioners may need to prioritize one fairness notion or adjust the Lagrange multiplier weights $\lambda_{\text{DP}}, \lambda_{\text{IF}}$.

Radius Selection Ablation We empirically validate the importance of using calibrated radii (Remark 6.2) by comparing DRO-FAIR with calibrated radii versus overly large radii ($\rho_{\text{DP},j} = \rho_{\text{IF}} = 1$) in Table 5. Using overly large radii makes the worst-case constraints so conservative that the optimization struggles to satisfy them, effectively ignoring the fairness constraints. This results in (1) higher accuracy (constraints are not enforced) and (2) substantially worse fairness: DP violations are $2.5\times$ to $8\times$ higher, and IF violations are $3\times$ higher compared to calibrated radii. Conversely, as $\rho_{\text{DP},j}, \rho_{\text{IF}} \rightarrow 0$, the uncertainty sets shrink to point masses at the empirical distribution, and DRO-FAIR recovers the Naive-FAIR approach (no robustness). The calibrated radii (Theorems 4.2–4.3) balance this trade-off: they are large enough to contain the clean distribution $\mathbb{P}$ within the uncertainty set, yet small enough to yield meaningful worst-case constraints.

8. Conclusion

We presented DRO-FAIR, a DRO framework for joint DP and IF under α -corruption. Our tight TV bounds and calibrated uncertainty radii guarantee clean-distribution fairness, addressing corruption’s disproportionate harm to minority groups. Empirically, DRO-FAIR reduces DP violations by 83% and IF by 3–5 $\times$ at 1–4% accuracy cost. Future work may reduce overhead and improve noise estimation.

Impact Statement

This work addresses algorithmic fairness under data corruption for high-stakes domains such as healthcare, criminal justice, and lending. **Positive impacts:** DRO-FAIR provides certified fairness guarantees under the α -corruption model, helping prevent discriminatory outcomes from corrupted training data. Our analysis reveals that minority groups are disproportionately harmed by corruption (Theorem 4.2), and our corruption-calibrated uncertainty sets explicitly account for this vulnerability. **Potential risks:** Guarantees require accurate estimation of α ; underestimation may cause fairness violations to exceed bounds, while overestimation yields overly conservative classifiers. The $12\times$ computational overhead may limit resource-constrained deployments, and our random corruption experiments provide a lower bound-adversarial corruption may cause larger degradation. **Broader context:** No fairness intervention addresses all societal bias; algorithmic constraints cannot rectify upstream issues such as historical discrimination in labels. Our work is one component of a broader sociotechnical approach to fair AI.

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A. Empirical Formulas for Fairness Violations

For completeness, we provide the explicit formulas for the empirical fairness violations used throughout the paper.

Empirical Demographic Parity Violation. Given a sample $S = \{(x_i, a_i, y_i)\}_{i=1}^n$, let $n_a = \sum_{i=1}^n \mathbb{1}\{a_i = a\}$ denote the number of samples in group $a \in \{0, 1\}$. The empirical group-conditional positive rates are:

$$\hat{p}_a(h) = \frac{1}{n_a} \sum_{i: a_i = a} \mathbb{1}\{h(x_i) = 1\}. \quad (24)$$

The empirical Demographic Parity Violation is then:

$$\hat{\Delta}_{\text{DP}}(h; S) = |\hat{p}_0(h) - \hat{p}_1(h)|. \quad (25)$$

Empirical Individual Fairness Violation. The empirical Individual Fairness Violation is an U-statistic of order 2:

$$\hat{L}_{d,\gamma}(h; S) = \frac{1}{n(n-1)} \sum_{i \neq j} \mathbb{1}(|h(x_i) - h(x_j)| > d(x_i, x_j) + \gamma), \quad (26)$$

where $d : \mathcal{X} \times \mathcal{X} \rightarrow [0, 1]$ is the task-specific similarity metric and $\gamma \geq 0$ is the slack parameter.

B. Related Work Comparison

Table 6. Comparison of fairness approaches. Corruption types: NS = natural shift, MN = malicious noise (α -corruption). Type: algorithm (DRO, Robust Tr.) vs. hardness.

Approach	Group	Individual	Corruption	Type
Hashimoto et al. '18	✓	×	NS	DRO
Sagawa et al. '20	✓	×	NS	DRO
Levy et al. '20	✓	×	NS	DRO
Diana et al. '21	✓	×	NS	Minimax
Wang et al. '23	✓	×	NS	Robust
Ehyaie et al. '24	×	✓	NS	Causal DRO
S. Wang et al. '20	✓	×	MN	DRO
Roh et al. '20	✓	×	MN	Defense
Konstantinov & Lampert '22	✓	×	MN	Hardness
Blum et al. '24	✓	×	MN	Hardness
This work	✓	✓	MN	Constructive DRO

C. Proofs for Section 4: Fairness under the α -Corruption Model

C.1. Proof of Theorem 4.2

Proof. By Section 3.1, we have $\pi_j = \mathbb{P}(A = j) > 0$ for both groups, ensuring all conditional distributions are well-defined.

Fix a group $j \in \{0, 1\}$. We derive a bound on the total variation (TV) distance between the clean and perturbed group-conditional distributions by expressing the perturbed conditional as an explicit mixture and applying TV convexity.

Notation. Define:

$$\begin{aligned} \pi_j &:= \mathbb{P}(A = j), & q_j &:= \mathbb{Q}(A = j) \quad (\text{group marginals}), \\ \mathbb{P}_j &:= \mathbb{P}((X, Y) \mid A = j), & \mathbb{Q}_j &:= \mathbb{Q}((X, Y) \mid A = j) \quad (\text{group conditionals}), \\ \mathbb{P}_{\text{pert},j} &:= \mathbb{P}_{\text{pert}}((X, Y) \mid A = j) \quad (\text{perturbed group conditional}). \end{aligned}$$

Note that $\mathbb{P}_{\text{pert}}(A = j) = (1 - \alpha)\pi_j + \alpha q_j \geq (1 - \alpha)\pi_j > 0$, so $\mathbb{P}_{\text{pert},j}$ is well-defined.

Step 1: Mixture representation. Under Definition 4.1, the perturbed distribution is $\mathbb{P}_{\text{pert}} = (1 - \alpha)\mathbb{P} + \alpha\mathbb{Q}$. By the definition of conditional probability and linearity of measures,

$$\begin{aligned}\mathbb{P}_{\text{pert},j}(B) &= \frac{\mathbb{P}_{\text{pert}}((X, Y) \in B, A = j)}{\mathbb{P}_{\text{pert}}(A = j)} \\ &= \frac{(1 - \alpha)\mathbb{P}((X, Y) \in B, A = j) + \alpha\mathbb{Q}((X, Y) \in B, A = j)}{(1 - \alpha)\pi_j + \alpha q_j} \\ &= \frac{(1 - \alpha)\pi_j \mathbb{P}_j(B) + \alpha q_j \mathbb{Q}_j(B)}{(1 - \alpha)\pi_j + \alpha q_j}.\end{aligned}$$

Define the mixture weight

$$w := \frac{(1 - \alpha)\pi_j}{(1 - \alpha)\pi_j + \alpha q_j} \in [0, 1].$$

Then $\mathbb{P}_{\text{pert},j} = w\mathbb{P}_j + (1 - w)\mathbb{Q}_j$, where $1 - w = \frac{\alpha q_j}{(1 - \alpha)\pi_j + \alpha q_j}$.

Step 2: Bounding the total variation distance. By the convexity property of total variation, for any distributions μ, ν and $a \in [0, 1]$,

$$\text{TV}(\mu, a\mu + (1 - a)\nu) = (1 - a) \text{TV}(\mu, \nu).$$

Applying this with $\mu = \mathbb{P}_j$, $\nu = \mathbb{Q}_j$, and $a = w$:

$$\text{TV}(\mathbb{P}_j, \mathbb{P}_{\text{pert},j}) = (1 - w) \text{TV}(\mathbb{P}_j, \mathbb{Q}_j) \leq 1 - w = \frac{\alpha q_j}{(1 - \alpha)\pi_j + \alpha q_j},$$

where the inequality uses $\text{TV}(\mathbb{P}_j, \mathbb{Q}_j) \leq 1$.

Step 3: Worst-case analysis. To bound the corruption effect in the worst case, we maximize over $q_j \in [0, 1]$. The function

$$f(q_j) = \frac{\alpha q_j}{(1 - \alpha)\pi_j + \alpha q_j}$$

has derivative

$$\frac{df}{dq_j} = \frac{\alpha(1 - \alpha)\pi_j}{((1 - \alpha)\pi_j + \alpha q_j)^2} > 0 \quad \text{for } \pi_j > 0,$$

hence f is strictly increasing on $[0, 1]$. The maximum occurs at $q_j = 1$, yielding

$$\text{TV}(\mathbb{P}_{X,Y|A=j}, \mathbb{P}_{\text{pert},X,Y|A=j}) \leq \frac{\alpha}{(1 - \alpha)\pi_j + \alpha} \leq 1.$$

This completes the proof. $\square$

C.2. Proof of Theorem 4.3

Proof. Let $\mathbb{P}_X$ and $\mathbb{Q}_X$ be the marginal distributions on $\mathcal{X}$. Under the α -corruption model, the perturbed marginal distribution is:

$$\mathbb{P}_{\text{pert},X} = (1 - \alpha)\mathbb{P}_X + \alpha\mathbb{Q}_X.$$

We bound the total variation distance via a coupling argument. Consider two independent draws from $\mathbb{P}_{\text{pert},X}$: for each draw, with probability $(1 - \alpha)$ it comes from $\mathbb{P}_X$, and with probability α it comes from $\mathbb{Q}_X$, independently.

Construct a coupling between $(X_1, X_2) \sim \mathbb{P}_X^{\otimes 2}$ and $(Z_1, Z_2) \sim \mathbb{P}_{\text{pert},X}^{\otimes 2}$ as follows:

- With probability $(1 - \alpha)^2$: set $(Z_1, Z_2) = (X_1, X_2)$.
- With probability $\alpha(1 - \alpha)$: sample $Z_1 \sim \mathbb{Q}_X$ independently, set $Z_2 = X_2$.
- With probability $\alpha(1 - \alpha)$: set $Z_1 = X_1$, sample $Z_2 \sim \mathbb{Q}_X$ independently.

- With probability α^2 : sample $(Z_1, Z_2) \sim \mathbb{Q}_X^{\otimes 2}$ independently.

By construction, Z_1 and Z_2 are independent, and each Z_i follows $\mathbb{P}_{\text{pert}, X}$ marginally. Thus (Z_1, Z_2) has the correct distribution $\mathbb{P}_{\text{pert}, X}^{\otimes 2}$.

The pairs (X_1, X_2) and (Z_1, Z_2) are equal only in the first case. Therefore,

$$\Pr((X_1, X_2) \neq (Z_1, Z_2)) = 1 - (1 - \alpha)^2 = 2\alpha - \alpha^2.$$

By the coupling characterization of total variation distance,

$$\text{TV}(\mathbb{P}_X^{\otimes 2}, \mathbb{P}_{\text{pert}, X}^{\otimes 2}) \leq \Pr((X_1, X_2) \neq (Z_1, Z_2)) = 2\alpha - \alpha^2.$$

The bound is tight: when $\mathbb{P}_X$ and $\mathbb{Q}_X$ have disjoint supports, the probability of (Z_1, Z_2) falling in the support of $\mathbb{P}_X^{\otimes 2}$ is exactly $(1 - \alpha)^2$, giving

$$\text{TV}(\mathbb{P}_X^{\otimes 2}, \mathbb{P}_{\text{pert}, X}^{\otimes 2}) = 1 - (1 - \alpha)^2 = 2\alpha - \alpha^2.$$

□

C.3. Proof of Corollary 4.4

Proof. Let $B = \{(x, y) \in \mathcal{X} \times \mathcal{Y} : h(x) = 1\}$. The difference in group-conditional acceptance rates is

$$|\mathbb{P}(h(X) = 1 \mid A = j) - \mathbb{P}_{\text{pert}}(h(X) = 1 \mid A = j)| = |\mathbb{P}_j(B) - \mathbb{P}_{\text{pert}, j}(B)|.$$

By Theorem 4.2, the perturbed conditional has the mixture representation $\mathbb{P}_{\text{pert}, j} = w\mathbb{P}_j + (1 - w)\mathbb{Q}_j$ with $w = \frac{(1 - \alpha)\pi_j}{(1 - \alpha)\pi_j + \alpha q_j}$. For any measurable set B ,

$$|\mathbb{P}_j(B) - \mathbb{P}_{\text{pert}, j}(B)| = |\mathbb{P}_j(B) - w\mathbb{P}_j(B) - (1 - w)\mathbb{Q}_j(B)| = (1 - w) |\mathbb{P}_j(B) - \mathbb{Q}_j(B)| \leq 1 - w.$$

Maximizing over $q_j \in [0, 1]$ (maximum at $q_j = 1$ by Theorem 4.2) yields

$$|\mathbb{P}(h(X) = 1 \mid A = j) - \mathbb{P}_{\text{pert}}(h(X) = 1 \mid A = j)| \leq \frac{\alpha}{(1 - \alpha)\pi_j + \alpha}.$$

□

C.4. Proof of Corollary 4.5

Proof. Define the fairness violation indicator:

$$\phi(x, x') = \mathbb{1}(|h(x) - h(x')| > d(x, x') + \gamma).$$

By definition,

$$L_{d, \gamma}(h; \mathbb{P}) = \mathbb{E}_{(X, X') \sim \mathbb{P}_X^{\otimes 2}}[\phi(X, X')].$$

Since h is binary, $|h(x) - h(x')| \in \{0, 1\}$, and $d(x, x') \geq 0$, we have $\phi(x, x') \in \{0, 1\} \subset [0, 1]$.

The difference in losses satisfies:

$$|L_{d, \gamma}(h; \mathbb{P}) - L_{d, \gamma}(h; \mathbb{P}_{\text{pert}})| = \left| \mathbb{E}_{\mathbb{P}_X^{\otimes 2}}[\phi] - \mathbb{E}_{\mathbb{P}_{\text{pert}, X}^{\otimes 2}}[\phi] \right|.$$

For any distributions μ, ν and any function ψ with $\|\psi\|_{\infty} \leq 1$,

$$|\mathbb{E}_{\mu}[\psi] - \mathbb{E}_{\nu}[\psi]| \leq \text{TV}(\mu, \nu).$$

Applying this with $\mu = \mathbb{P}_X^{\otimes 2}$, $\nu = \mathbb{P}_{\text{pert},X}^{\otimes 2}$, and $\psi = \phi$:

$$|L_{d,\gamma}(h; \mathbb{P}) - L_{d,\gamma}(h; \mathbb{P}_{\text{pert}})| \leq \text{TV}(\mathbb{P}_X^{\otimes 2}, \mathbb{P}_{\text{pert},X}^{\otimes 2}).$$

By Theorem 4.3,

$$\text{TV}(\mathbb{P}_X^{\otimes 2}, \mathbb{P}_{\text{pert},X}^{\otimes 2}) \leq 2\alpha - \alpha^2,$$

which completes the proof. $\square$

D. Proofs for Section 5: Naive-FAIR: Naive Approach

D.1. Supporting Lemmas

We begin by stating the three key lemmas that bound the uniform deviation of each component.

Lemma D.1 (Uniform Convergence for Classification Loss). *Let $\mathcal{H}$ have VC dimension D_{VC} . Under the α -corruption model (Definition 4.1), let $S^c = \{(x_i^c, a_i^c, y_i^c)\}_{i=1}^n$ denote the corrupted sample with marginal distribution $\mathbb{P}_{\text{pert}} = (1 - \alpha)\mathbb{P} + \alpha\mathbb{Q}$. If $n \geq 10 \log(6/\delta)/\alpha$, then with probability at least $1 - \delta/3$,*

$$\sup_{h \in \mathcal{H}} |\hat{R}(h; S^c) - R(h; \mathbb{P})| \leq \frac{3\alpha}{2} + 2\sqrt{\frac{8D_{\text{VC}} \log(en/D_{\text{VC}}) + 2 \log(24/\delta)}{n}}.$$

Lemma D.2 (Uniform Convergence for Individual Fairness Loss). *Under the same setting as Lemma D.1, with probability at least $1 - \delta/3$,*

$$\sup_{h \in \mathcal{H}} |\hat{L}_{d,\gamma}(h; S^c) - L_{d,\gamma}(h; \mathbb{P})| \leq (2\alpha - \alpha^2) + 2\sqrt{\frac{16D_{\text{VC}} \log(en/(2D_{\text{VC}})) + 2 \log(12/\delta)}{n}}.$$

Lemma D.3 (Uniform Convergence for Demographic Parity Loss). *Under the same setting as Lemma D.1, if additionally*

$$n \geq \max \left\{ \frac{8 \log(12/\delta)}{(1 - \alpha)\pi_0}, \frac{2D_{\text{VC}}}{(1 - \alpha)\pi_0} \right\},$$

then with probability at least $1 - \delta/3$,

$$\sup_{h \in \mathcal{H}} |\hat{\Delta}_{\text{DP}}(h; S^c) - \Delta_{\text{DP}}(h; \mathbb{P})| \leq \frac{2\alpha}{(1 - \alpha)\pi_0 + \alpha} + 4\sqrt{2} \sqrt{\frac{8D_{\text{VC}} \log(en/D_{\text{VC}}) + 2 \log(48/\delta)}{(1 - \alpha)\pi_0 n}}.$$

We now provide proof sketches for each lemma.

D.2. Proof of Lemma D.1

Proof. Recall that $S = \{(x_i, a_i, y_i)\}_{i=1}^n$ denotes the unobserved clean sample drawn i.i.d. from $\mathbb{P}$, and $S^c = \{(x_i^c, a_i^c, y_i^c)\}_{i=1}^n$ denotes the observed corrupted sample drawn i.i.d. from $\mathbb{P}_{\text{pert}}$. We aim to bound the uniform deviation

$$\sup_{h \in \mathcal{H}} |\hat{R}(h; S^c) - R(h; \mathbb{P})|.$$

The key insight is to use the latent clean sample S as an intermediary: although S is unobserved, we can reason about it probabilistically since its distribution is known.

By the triangle inequality,

$$|\hat{R}(h; S^c) - R(h; \mathbb{P})| \leq \underbrace{|\hat{R}(h; S^c) - \hat{R}(h; S)|}_{\text{corruption effect}} + \underbrace{|\hat{R}(h; S) - R(h; \mathbb{P})|}_{\text{sampling error}}$$

and taking supremum over $h \in \mathcal{H}$:

$$\sup_{h \in \mathcal{H}} |\hat{R}(h; S^c) - R(h; \mathbb{P})| \leq \sup_{h \in \mathcal{H}} |\hat{R}(h; S^c) - \hat{R}(h; S)| + \sup_{h \in \mathcal{H}} |\hat{R}(h; S) - R(h; \mathbb{P})|.$$

We bound each term separately and combine via the union bound.

1. Clean Data Deviation (Sampling Error). Since $S \sim \mathbb{P}$ is i.i.d., we apply Theorem 3.3 with failure probability $\delta' = \delta/6$. With probability at least $1 - \delta/6$ over the randomness of S ,

$$\sup_{h \in \mathcal{H}} |\widehat{R}(h; S) - R(h; \mathbb{P})| \leq 2 \sqrt{\frac{8D_{\text{VC}} \log\left(\frac{en}{D_{\text{VC}}}\right) + 2 \log\left(\frac{24}{\delta}\right)}{n}},$$

where we used $\log(4/\delta') = \log(4 \cdot 6/\delta) = \log(24/\delta)$.

2. Corruption Effect. Each example in S is independently selected for corruption with probability α to yield S^c (Definition 4.1). Let $M \subseteq [n]$ denote the random set of corrupted indices, so $|M| \sim \text{Bin}(n, \alpha)$ with $\mathbb{E}[|M|] = \alpha n$. For $i \notin M$, we have $(x_i^c, a_i^c, y_i^c) = (x_i, a_i, y_i)$, so the 0-1 loss terms cancel; for $i \in M$, the per-sample difference is at most 1. Thus for any fixed h ,

$$|\widehat{R}(h; S^c) - \widehat{R}(h; S)| = \left| \frac{1}{n} \sum_{i \in M} (\mathbb{1}(h(x_i^c) \neq y_i^c) - \mathbb{1}(h(x_i) \neq y_i)) \right| \leq \frac{|M|}{n},$$

and hence (since the bound is hypothesis-independent):

$$\sup_{h \in \mathcal{H}} |\widehat{R}(h; S^c) - \widehat{R}(h; S)| \leq \frac{|M|}{n}.$$

We now apply the multiplicative Chernoff bound. For $X \sim \text{Bin}(n, p)$ with mean $\mu = np$, the multiplicative form (Mitzenmacher & Upfal (2005)) states:

$$\Pr(X \geq (1 + \delta')\mu) \leq \exp\left(-\frac{(\delta')^2 \mu}{2 + \delta'}\right).$$

Setting $\delta' = 1/2$ (so $(1 + \delta')\mu = 1.5\alpha n$) with $X = |M|$ and $\mu = \alpha n$:

$$\Pr(|M| \geq \frac{3\alpha n}{2}) \leq \exp\left(-\frac{(1/2)^2 \cdot \alpha n}{2 + 1/2}\right) = \exp\left(-\frac{0.25 \cdot \alpha n}{2.5}\right) = \exp\left(-\frac{\alpha n}{10}\right).$$

Set $n \geq 10 \log(6/\delta)/\alpha$; then $\exp(-\alpha n/10) \leq \exp(-\log(6/\delta)) = \delta/6$. Thus with probability at least $1 - \delta/6$ over the randomness of corruption, $|M| \leq 3\alpha n/2$, which implies:

$$\sup_{h \in \mathcal{H}} |\widehat{R}(h; S^c) - \widehat{R}(h; S)| \leq \frac{3\alpha}{2}.$$

3. Combine with Union Bound. The clean data deviation event fails with probability at most $\delta/6$, and the corruption concentration event fails with probability at most $\delta/6$. By a union bound, both events hold simultaneously with probability at least $1 - \delta/6 - \delta/6 = 1 - \delta/3$.

Thus, with probability at least $1 - \delta/3$ over the joint randomness of sampling S from $\mathbb{P}$ and corrupting S to obtain S^c :

$$\sup_{h \in \mathcal{H}} |\widehat{R}(h; S^c) - R(h; \mathbb{P})| \leq \frac{3\alpha}{2} + 2 \sqrt{\frac{8D_{\text{VC}} \log\left(\frac{en}{D_{\text{VC}}}\right) + 2 \log\left(\frac{24}{\delta}\right)}{n}}.$$

□

D.3. Proof of Lemma D.2

Proof. Define the pairwise violation kernel

$$\phi_h(x, x') := \mathbb{1}\{|h(x) - h(x')| > d(x, x') + \gamma\}.$$

The corrupted samples $S^c = \{(x_i^c, a_i^c, y_i^c)\}_{i=1}^n$ are i.i.d. from $\mathbb{P}_{\text{pert}}$, so pairs (x_i^c, x_j^c) for $i \neq j$ are drawn from $\mathbb{P}_{\text{pert}, X}^{\otimes 2}$. By the triangle inequality we have

$$|\widehat{L}_{d, \gamma}(h; S^c) - L_{d, \gamma}(h; \mathbb{P})| \leq \underbrace{|\widehat{L}_{d, \gamma}(h; S^c) - L_{d, \gamma}(h; \mathbb{P}_{\text{pert}})|}_{\text{sampling error}} + \underbrace{|L_{d, \gamma}(h; \mathbb{P}_{\text{pert}}) - L_{d, \gamma}(h; \mathbb{P})|}_{\text{corruption bias}}.$$

Corruption bias. By Corollary 4.5 we have, for all $h \in \mathcal{H}$,

$$|L_{d,\gamma}(h; \mathbb{P}_{\text{pert}}) - L_{d,\gamma}(h; \mathbb{P})| \leq \text{TV}(\mathbb{P}_X^{\otimes 2}, \mathbb{P}_{\text{pert},X}^{\otimes 2}) \leq 2\alpha - \alpha^2,$$

so the corruption bias is deterministic and bounded by $2\alpha - \alpha^2$.

Sampling error. The empirical loss

$$\hat{L}_{d,\gamma}(h; S^c) = \frac{1}{n(n-1)} \sum_{i \neq j} \phi_h(x_i^c, x_j^c)$$

is a U-statistic of order 2. We establish the VC dimension of the kernel class and apply U-process concentration theory.

VC dimension of the kernel class. Let $\mathcal{F} := \{\phi_h : h \in \mathcal{H}\}$. For binary classifiers $h : \mathcal{X} \rightarrow \{0, 1\}$ the condition $|h(x) - h(x')| > d(x, x') + \gamma$ (with $d \in [0, 1]$, $\gamma \geq 0$) is equivalent to

$$h(x) \neq h(x') \quad \text{and} \quad d(x, x') + \gamma < 1.$$

Hence

$$\phi_h(x, x') = \mathbb{1}\{h(x) \neq h(x')\} \cdot \mathbb{1}\{d(x, x') + \gamma < 1\}.$$

If $\gamma \geq 1$ then $\mathcal{F} = \{0\}$ and $\text{VC}(\mathcal{F}) = 0$. Otherwise ϕ_h depends on the pairwise XOR $h(x) \oplus h(x')$ restricted to “close” pairs. Define the pairwise class

$$\mathcal{G} := \{(x, x') \mapsto h(x) \oplus h(x') : h \in \mathcal{H}\}.$$

Since restriction cannot increase VC dimension, $\text{VC}(\mathcal{F}) \leq \text{VC}(\mathcal{G})$. By standard results on pairwise evaluation of a single Boolean function (Blumer et al., 1989) we have $\text{VC}(\mathcal{G}) \leq 2D_{\text{VC}}$. Hence

$$\text{VC}(\mathcal{F}) \leq 2D_{\text{VC}}.$$

U-process concentration. For U-statistics of order 2 with kernel class $\mathcal{F}$ of VC dimension $D_{\mathcal{F}} \leq 2D_{\text{VC}}$, established U-process uniform convergence results (Cléménçon et al., 2008; Arcones & Giné, 1993) yield that with probability at least $1 - \delta/3$,

$$\sup_{h \in \mathcal{H}} |\hat{L}_{d,\gamma}(h; S^c) - L_{d,\gamma}(h; \mathbb{P}_{\text{pert}})| \leq C \sqrt{\frac{D_{\mathcal{F}}(\log n + \log(3/\delta))}{n}},$$

where C is a constant depending on the U-process bounds for order-2 kernels. Adapting constants to match the form of Theorem 3.3 and using $D_{\mathcal{F}} \leq 2D_{\text{VC}}$ gives:

$$\sup_{h \in \mathcal{H}} |\hat{L}_{d,\gamma}(h; S^c) - L_{d,\gamma}(h; \mathbb{P}_{\text{pert}})| \leq 2 \sqrt{\frac{16D_{\text{VC}} \log(\frac{en}{2D_{\text{VC}}}) + 2 \log(12/\delta)}{n}}.$$

Combining. The corruption bias bound is deterministic and the sampling error bound holds with probability at least $1 - \delta/3$. Therefore, with probability at least $1 - \delta/3$,

$$\sup_{h \in \mathcal{H}} |\hat{L}_{d,\gamma}(h; S^c) - L_{d,\gamma}(h; \mathbb{P})| \leq (2\alpha - \alpha^2) + 2 \sqrt{\frac{16D_{\text{VC}} \log(\frac{en}{2D_{\text{VC}}}) + 2 \log(12/\delta)}{n}}.$$

This completes the proof. $\square$

Remark D.4. The VC dimension bound $\text{VC}(\mathcal{G}) \leq 2D_{\text{VC}}$ applies to pairwise evaluation of a single function on two inputs, as opposed to symmetric differences of different functions (which can have $\Omega(D_{\text{VC}} \log D_{\text{VC}})$ dimension). The U-process constant above is chosen for comparability with Theorem 3.3; rigorous U-process theory guarantees the stated rate, possibly with larger absolute constants.

D.4. Proof of Lemma D.3

Proof. We bound $\sup_{h \in \mathcal{H}} |\hat{\Delta}_{\text{DP}}(h; S^c) - \Delta_{\text{DP}}(h; \mathbb{P})|$ by decomposing into corruption bias and sampling error. Throughout, we assume $\pi_0 > 0$ (non-empty minority group).

Notation. For $a \in \{0, 1\}$ and $h \in \mathcal{H}$, let $p_a(h) = \mathbb{P}(h(X) = 1 \mid A = a)$, $p_a^{\text{pert}}(h) = \mathbb{P}_{\text{pert}}(h(X) = 1 \mid A = a)$, and $\hat{p}_a(h) = \frac{1}{n_a^c} \sum_{i: a_i^c = a} \mathbb{1}\{h(x_i^c) = 1\}$, where $n_a^c = \sum_{i=1}^n \mathbb{1}\{a_i^c = a\}$. Then $\Delta_{\text{DP}}(h; \mathbb{P}) = |p_0(h) - p_1(h)|$ and $\hat{\Delta}_{\text{DP}}(h; S^c) = |\hat{p}_0(h) - \hat{p}_1(h)|$.

Decomposition. By the reverse triangle inequality $||a - b| - |c - d|| \leq |a - c| + |b - d|$:

$$\sup_{h \in \mathcal{H}} |\hat{\Delta}_{\text{DP}}(h; S^c) - \Delta_{\text{DP}}(h; \mathbb{P})| \leq \sum_{a \in \{0, 1\}} \sup_{h \in \mathcal{H}} \underbrace{|\hat{p}_a(h) - p_a(h)|}_{\text{per-group error}}. \quad (27)$$

Each per-group error decomposes as:

$$|\hat{p}_a(h) - p_a(h)| \leq \underbrace{|\hat{p}_a(h) - p_a^{\text{pert}}(h)|}_{\text{sampling error}} + \underbrace{|p_a^{\text{pert}}(h) - p_a(h)|}_{\text{corruption bias}}.$$

Corruption bias. By Corollary 4.4, for each $a \in \{0, 1\}$:

$$\sup_{h \in \mathcal{H}} |p_a^{\text{pert}}(h) - p_a(h)| \leq \frac{\alpha}{(1 - \alpha)\pi_a + \alpha}.$$

This is deterministic and holds for both groups.

Sampling error. The samples with $a_i^c = a$ are i.i.d. from $\mathbb{P}_{\text{pert}}(\cdot \mid A = a)$ with random size n_a^c . We condition on n_a^c being sufficiently large and apply VC uniform convergence.

Step 1: Group size concentration.

We ensure each group has sufficient samples with high probability. Let $E_a = \{n_a^c \geq (1 - \alpha)\pi_a n / 2\}$ denote the event that group a has at least half its worst-case expected size. The expectation of n_a^c satisfies:

$$\mathbb{E}[n_a^c] = n\mathbb{P}_{\text{pert}}(A = a) = n((1 - \alpha)\pi_a + \alpha q_a) \geq (1 - \alpha)\pi_a n,$$

where $q_a = \mathbb{Q}(A = a) \geq 0$ is the corruption distribution's group assignment probability. Since $n_a^c = \sum_{i=1}^n \mathbb{1}\{a_i^c = a\}$ is a sum of independent Bernoulli random variables, the multiplicative Chernoff bound (lower tail, $\delta' = 1/2$) gives:

$$\Pr(E_a^c) = \Pr\left(n_a^c \leq \frac{(1 - \alpha)\pi_a n}{2}\right) \leq \Pr\left(n_a^c \leq \frac{\mathbb{E}[n_a^c]}{2}\right) \leq \exp\left(-\frac{\mathbb{E}[n_a^c]}{8}\right) \leq \exp\left(-\frac{(1 - \alpha)\pi_a n}{8}\right).$$

For this to be at most $\delta/12$, we require $(1 - \alpha)\pi_a n / 8 \geq \log(12/\delta)$, giving the first condition below. The second condition ensures that on E_a , the group sample size $(1 - \alpha)\pi_a n / 2$ exceeds D_{VC} , making the VC bound in Step 2 non-vacuous. Since $\pi_0 \leq \pi_1$, the minority group $a = 0$ determines both requirements:

$$n \geq \max\left\{\frac{8 \log(12/\delta)}{(1 - \alpha)\pi_0}, \frac{2D_{\text{VC}}}{(1 - \alpha)\pi_0}\right\}.$$

Step 2: VC uniform convergence.

We apply VC theory conditionally on the group size being large enough. The key is that restricting to group a does not increase complexity.

VC dimension of restricted class. Define $\mathcal{G}_a = \{(x, a') \mapsto \mathbb{1}\{h(x) = 1, a' = a\} : h \in \mathcal{H}\}$. We claim $\text{VCdim}(\mathcal{G}_a) \leq D_{\text{VC}}$. To see this, suppose $\mathcal{G}_a$ shatters a set $S = \{(x_i, a'_i)\}_{i=1}^m$. For any (x_i, a'_i) with $a'_i \neq a$, the function output is always 0 regardless of h , so the labeling $y_i = 1$ is unachievable. Thus all shattered points must have $a'_i = a$, reducing to shattering $\{x_i\}$ by $\mathcal{H}$. Hence $\text{VCdim}(\mathcal{G}_a) \leq D_{\text{VC}}$.

Conditional uniform convergence. For any fixed sample size $m \geq (1 - \alpha)\pi_a n / 2 \geq D_{\text{VC}}$, conditioning on $n_a^c = m$ makes the group- a samples i.i.d. from $\mathbb{P}_{\text{pert}}(\cdot \mid A = a)$. Applying Theorem 3.3 to these m samples with VC dimension D_{VC} and failure probability $\delta/12$:

$$\Pr\left(\sup_{h \in \mathcal{H}} |\hat{p}_a(h) - p_a^{\text{pert}}(h)| > \varepsilon_a(m) \mid n_a^c = m\right) \leq \frac{\delta}{12},$$

where $\varepsilon_a(m) = 2\sqrt{\frac{8D_{\text{VC}} \log(em/D_{\text{VC}}) + 2\log(48/\delta)}{m}}$.

Simplifying the bound.

By the law of total probability over all possible values of n_a^c , the VC bound holds unconditionally with probability $\geq 1 - \delta/12$. Let F_a denote the event that this VC bound holds for group a . To obtain an explicit non-random bound, we use:

- **In the logarithm:** $n_a^c \leq n$ (conservative, always true) gives $\log(en_a^c/D_{\text{VC}}) \leq \log(en/D_{\text{VC}})$.
- **In the denominator:** On E_a , we have $n_a^c \geq (1 - \alpha)\pi_a n/2$, so $1/n_a^c \leq 2/((1 - \alpha)\pi_a n)$.

Combining these on $E_a \cap F_a$:

$$\sup_{h \in \mathcal{H}} |\hat{p}_a(h) - p_a^{\text{pert}}(h)| \leq 2\sqrt{\frac{8D_{\text{VC}} \log(en/D_{\text{VC}}) + 2\log(48/\delta)}{(1 - \alpha)\pi_a n/2}} = 2\sqrt{2}\sqrt{\frac{8D_{\text{VC}} \log(en/D_{\text{VC}}) + 2\log(48/\delta)}{(1 - \alpha)\pi_a n}}.$$

This simplified bound holds on $E_a \cap F_a$, which has probability $\geq 1 - \delta/6$ (by union bound: $\Pr(E_a^c) \leq \delta/12$ and $\Pr(F_a^c) \leq \delta/12$, so $\Pr(E_a^c \cup F_a^c) \leq \delta/12 + \delta/12 = \delta/6$).

Union bound. We have four events $\{E_0, F_0, E_1, F_1\}$, each failing with probability $\leq \delta/12$. By union bound, $\Pr(E_0 \cap F_0 \cap E_1 \cap F_1) \geq 1 - 4 \cdot \delta/12 = 1 - \delta/3$.

Final bound. Summing over $a \in \{0, 1\}$ and using $\pi_0 \leq \pi_1$ (so the worst-case group is $a = 0$):

$$\sum_{a \in \{0, 1\}} \frac{\alpha}{(1 - \alpha)\pi_a + \alpha} \leq \frac{2\alpha}{(1 - \alpha)\pi_0 + \alpha},$$

$$\sum_{a \in \{0, 1\}} 2\sqrt{2}\sqrt{\frac{8D_{\text{VC}} \log(en/D_{\text{VC}}) + 2\log(48/\delta)}{(1 - \alpha)\pi_a n}} \leq 4\sqrt{2}\sqrt{\frac{8D_{\text{VC}} \log(en/D_{\text{VC}}) + 2\log(48/\delta)}{(1 - \alpha)\pi_0 n}}.$$

Combining gives the stated bound with probability $\geq 1 - \delta/3$. $\square$

Remark D.5. The conditioning argument in Step 2 is standard: Theorem 3.3 applies to any fixed sample size, and for each m , the conditional probability of failure is $\leq \delta/12$. Averaging over the random n_a^c (law of total probability) preserves this bound. The union bound accounts for both the concentration event E_a and the VC event F_a for each group.

D.5. Proof of Theorem 5.1

Proof. We combine Lemmas D.1–D.3 to bound the excess loss of the empirical minimizer

$$\hat{h} = \arg \min_{h \in \mathcal{H}} \hat{\mathcal{L}}_{\text{naive}}(h, \lambda_{\text{DP}}, \lambda_{\text{IF}}; S^c)$$

relative to the population minimizer

$$h^* = \arg \min_{h \in \mathcal{H}} \mathcal{L}_{\text{naive}}(h, \lambda_{\text{DP}}, \lambda_{\text{IF}}; \mathbb{P}).$$

Sample size requirement. The conditions from Lemmas D.1–D.3 require

$$n \geq \max \left\{ \frac{10 \log(6/\delta)}{\alpha}, \frac{8 \log(12/\delta)}{(1 - \alpha)\pi_0}, \frac{2D_{\text{VC}}}{(1 - \alpha)\pi_0} \right\}.$$

Bounding excess loss. Since $\hat{h}$ minimizes the empirical loss, $\hat{\mathcal{L}}_{\text{naive}}(\hat{h}) \leq \hat{\mathcal{L}}_{\text{naive}}(h^*)$. By standard symmetrization:

$$\mathcal{L}_{\text{naive}}(\hat{h}) - \mathcal{L}_{\text{naive}}(h^*) \leq 2 \sup_{h \in \mathcal{H}} |\mathcal{L}_{\text{naive}}(h) - \hat{\mathcal{L}}_{\text{naive}}(h)|.$$

Decomposition. By the triangle inequality:

$$\begin{aligned} \sup_{h \in \mathcal{H}} |\mathcal{L}_{\text{naive}}(h) - \hat{\mathcal{L}}_{\text{naive}}(h)| &\leq \sup_{h \in \mathcal{H}} |R(h; \mathbb{P}) - \hat{R}(h; S^c)| \\ &\quad + \lambda_{\text{IF}} \sup_{h \in \mathcal{H}} |L_{d,\gamma}(h; \mathbb{P}) - \hat{L}_{d,\gamma}(h; S^c)| \\ &\quad + \lambda_{\text{DP}} \sup_{h \in \mathcal{H}} |\Delta_{\text{DP}}(h; \mathbb{P}) - \hat{\Delta}_{\text{DP}}(h; S^c)|. \end{aligned}$$

Component bounds. Applying Lemmas D.1–D.3, each with failure probability $\delta/3$: *Classification loss (Lemma D.1):*

$$\sup_{h \in \mathcal{H}} |\hat{R}(h; S^c) - R(h; \mathbb{P})| \leq \frac{3\alpha}{2} + 2\sqrt{\frac{8D_{\text{VC}} \log(en/D_{\text{VC}}) + 2\log(24/\delta)}{n}}.$$

Metric fairness loss (Lemma D.2):

$$\sup_{h \in \mathcal{H}} |\hat{L}_{d,\gamma}(h; S^c) - L_{d,\gamma}(h; \mathbb{P})| \leq (2\alpha - \alpha^2) + 2\sqrt{\frac{16D_{\text{VC}} \log(en/(2D_{\text{VC}})) + 2\log(12/\delta)}{n}}.$$

Demographic parity loss (Lemma D.3):

$$\sup_{h \in \mathcal{H}} |\hat{\Delta}_{\text{DP}}(h; S^c) - \Delta_{\text{DP}}(h; \mathbb{P})| \leq \frac{2\alpha}{(1-\alpha)\pi_0 + \alpha} + 4\sqrt{2} \sqrt{\frac{8D_{\text{VC}} \log(en/D_{\text{VC}}) + 2\log(48/\delta)}{(1-\alpha)\pi_0 n}}.$$

Union bound and final result. By union bound over three events with failure probability $\delta/3$ each, all bounds hold simultaneously with probability $\geq 1 - \delta$. Substituting into the decomposition and applying the symmetrization factor of 2:

$$\mathcal{L}_{\text{naive}}(\hat{h}, \lambda_{\text{DP}}, \lambda_{\text{IF}}; \mathbb{P}) \leq \min_{h \in \mathcal{H}} \mathcal{L}_{\text{naive}}(h, \lambda_{\text{DP}}, \lambda_{\text{IF}}; \mathbb{P}) + \Delta_{\lambda},$$

where the excess loss satisfies

$$\begin{aligned} \Delta_{\lambda} &= 2 \left[\frac{3\alpha}{2} + \lambda_{\text{IF}}(2\alpha - \alpha^2) + \frac{2\lambda_{\text{DP}}\alpha}{(1-\alpha)\pi_0 + \alpha} \right] \\ &\quad + (1 + \lambda_{\text{IF}}) \cdot O\left(\sqrt{\frac{D_{\text{VC}} \log n + \log(1/\delta)}{n}}\right) + \lambda_{\text{DP}} \cdot O\left(\sqrt{\frac{D_{\text{VC}} \log n + \log(1/\delta)}{(1-\alpha)\pi_0 n}}\right). \end{aligned}$$

□

E. Auxiliary Lemmas for Robust Expectation

We now establish three auxiliary lemmas characterizing how expectations behave under ℓ_1 -ball (empirical) and TV-ball (population) perturbations. Lemmas E.1 and E.2 provide exact characterizations for binary classifiers, while Lemma E.4 extends these bounds to general bounded functions. These results are fundamental to the DRO analysis in the following sections.

Lemma E.1 (Robust Expectation under TV-Ball Perturbation). *Let $\mathbb{P}$ be a probability distribution over a measurable space $(\mathcal{X}, \mathcal{F})$, and let $h : \mathcal{X} \rightarrow \{0, 1\}$ be a measurable binary classifier. For any $\rho \geq 0$, define the Total-Variation (TV) ball uncertainty set*

$$\mathcal{B}_{\rho} = \{Q : \text{TV}(Q, \mathbb{P}) \leq \rho\},$$

where $\text{TV}(Q, \mathbb{P}) = \sup_{A \in \mathcal{F}} |Q(A) - \mathbb{P}(A)|$. Let $\mathbb{E}_{\mathbb{P}}[h] = \mathbb{P}(h = 1)$ denote the population mean.

Case 1 (Identically constant). *If h is identically constant on $\mathcal{X}$ (i.e., $h \equiv 0$ everywhere or $h \equiv 1$ everywhere), then*

$$\sup_{Q \in \mathcal{B}_{\rho}} \mathbb{E}_Q[h] = \inf_{Q \in \mathcal{B}_{\rho}} \mathbb{E}_Q[h] = c,$$

where $c \in \{0, 1\}$ is the constant value of h .

Case 2 (Non-constant). If both $\{x : h(x) = 0\}$ and $\{x : h(x) = 1\}$ are non-empty, then

$$\begin{aligned} \sup_{Q \in \mathcal{B}_\rho} \mathbb{E}_Q[h] &= \min(\mathbb{E}_\mathbb{P}[h] + \rho, 1), \\ \inf_{Q \in \mathcal{B}_\rho} \mathbb{E}_Q[h] &= \max(\mathbb{E}_\mathbb{P}[h] - \rho, 0). \end{aligned}$$

In particular, when h is $\mathbb{P}$ -almost surely constant but not identically constant ($\mathbb{E}_\mathbb{P}[h] \in \{0, 1\}$ but both level sets are non-empty):

- If $\mathbb{E}_\mathbb{P}[h] = 0$: $\sup = \min(\rho, 1)$ and $\inf = 0$.
- If $\mathbb{E}_\mathbb{P}[h] = 1$: $\sup = 1$ and $\inf = \max(1 - \rho, 0)$.

Proof of Lemma E.1. Let $A_1 = \{x \in \mathcal{X} : h(x) = 1\}$ and $A_0 = \{x \in \mathcal{X} : h(x) = 0\}$. Since h is measurable, both A_1 and A_0 are measurable sets with $A_1 \cup A_0 = \mathcal{X}$ and $A_1 \cap A_0 = \emptyset$. Recall that $\mathbb{E}_\mathbb{P}[h] = \mathbb{P}(A_1)$ as defined in the lemma statement.

Proof of Case 1 (Identically constant). If $h \equiv c$ for some $c \in \{0, 1\}$, then for any probability measure Q on $(\mathcal{X}, \mathcal{F})$:

$$\mathbb{E}_Q[h] = \int_{\mathcal{X}} h dQ = c \int_{\mathcal{X}} dQ = c.$$

Hence $\sup_{Q \in \mathcal{B}_\rho} \mathbb{E}_Q[h] = \inf_{Q \in \mathcal{B}_\rho} \mathbb{E}_Q[h] = c$.

Proof of Case 2 (Non-constant). Now assume both $A_1 \neq \emptyset$ and $A_0 \neq \emptyset$.

Step 1: Upper bound. For any $Q \in \mathcal{B}_\rho$, by the definition of TV distance applied to A_1 :

$$|Q(A_1) - \mathbb{P}(A_1)| \leq \text{TV}(Q, \mathbb{P}) \leq \rho.$$

Since $\mathbb{E}_Q[h] = Q(A_1)$, this gives $\mathbb{E}_Q[h] \leq \mathbb{E}_\mathbb{P}[h] + \rho$. Combined with $\mathbb{E}_Q[h] \leq 1$:

$$\sup_{Q \in \mathcal{B}_\rho} \mathbb{E}_Q[h] \leq \min(\mathbb{E}_\mathbb{P}[h] + \rho, 1).$$

Coupling interpretation. Alternatively, by the coupling characterization of TV distance, there exist random variables $X \sim \mathbb{P}$ and $Y \sim Q$ on a common probability space such that $\mathbb{P}(X \neq Y) \leq \rho$. Then $\mathbb{E}_Q[h] = \mathbb{E}[h(Y)] \leq \mathbb{E}[h(X)] + \mathbb{P}(X \neq Y) \leq \mathbb{E}_\mathbb{P}[h] + \rho$.

Step 2: Achieving the upper bound. We construct $Q^* \in \mathcal{B}_\rho$ attaining the bound. Since $A_1 \neq \emptyset$ and $A_0 \neq \emptyset$, we can define probability measures ν_1 on A_1 and ν_0 on A_0 :

- If $\mathbb{P}(A_1) > 0$: let $\nu_1(\cdot) = \mathbb{P}(\cdot | A_1)$.
- If $\mathbb{P}(A_1) = 0$: choose any $x_1 \in A_1$ and let $\nu_1 = \delta_{x_1}$ (the Dirac measure at x_1).
- Similarly for ν_0 on A_0 .

Subcase 2a: $\mathbb{E}_\mathbb{P}[h] + \rho \leq 1$. Define $Q^* = (\mathbb{E}_\mathbb{P}[h] + \rho) \nu_1 + (1 - \mathbb{E}_\mathbb{P}[h] - \rho) \nu_0$. Then:

$$\begin{aligned} Q^*(A_1) &= \mathbb{E}_\mathbb{P}[h] + \rho, \\ \text{TV}(Q^*, \mathbb{P}) &= |Q^*(A_1) - \mathbb{P}(A_1)| = \rho, \end{aligned}$$

where the first equality holds because $\{A_0, A_1\}$ partitions $\mathcal{X}$, so $|Q^*(A_0) - \mathbb{P}(A_0)| = |Q^*(A_1) - \mathbb{P}(A_1)|$, and this achieves the supremum in the TV definition.

Hence $Q^* \in \mathcal{B}_\rho$ and $\mathbb{E}_{Q^*}[h] = \mathbb{E}_\mathbb{P}[h] + \rho$.

Subcase 2b: $\mathbb{E}_\mathbb{P}[h] + \rho > 1$. Define $Q^* = \nu_1$. Then $Q^*(A_1) = 1$ and:

$$\text{TV}(Q^*, \mathbb{P}) = |Q^*(A_1) - \mathbb{P}(A_1)| = |1 - \mathbb{E}_\mathbb{P}[h]| = 1 - \mathbb{E}_\mathbb{P}[h] \leq \rho,$$

where the last inequality follows from the case condition $\mathbb{E}_\mathbb{P}[h] + \rho > 1$. Hence $Q^* \in \mathcal{B}_\rho$ and $\mathbb{E}_{Q^*}[h] = 1 = \min(\mathbb{E}_\mathbb{P}[h] + \rho, 1)$.

Combining: $\sup_{Q \in \mathcal{B}_\rho} \mathbb{E}_Q[h] = \min(\mathbb{E}_\mathbb{P}[h] + \rho, 1)$.

Step 3: Infimum via symmetry. Define $\tilde{h} = 1 - h$. Applying the supremum result to $\tilde{h}$:

$$\sup_{Q \in \mathcal{B}_\rho} \mathbb{E}_Q[\tilde{h}] = \min(1 - \mathbb{E}_\mathbb{P}[h] + \rho, 1).$$

Since $\sup_Q (1 - \mathbb{E}_Q[h]) = 1 - \inf_Q \mathbb{E}_Q[h]$:

$$\inf_{Q \in \mathcal{B}_\rho} \mathbb{E}_Q[h] = 1 - \min(1 - \mathbb{E}_\mathbb{P}[h] + \rho, 1) = \max(\mathbb{E}_\mathbb{P}[h] - \rho, 0).$$

Step 4: $\mathbb{P}$ -a.s. constant subcase. If $\mathbb{E}_\mathbb{P}[h] = 0$ but $A_1 \neq \emptyset$ (i.e., $h = 0$ $\mathbb{P}$ -a.s. but not identically 0): the formulas give $\sup = \min(\rho, 1)$ and $\inf = 0$. The supremum is achieved by the construction in Subcase 2a, taking $Q^* = \rho \nu_1 + (1 - \rho) \nu_0$, which moves mass ρ to A_1 .

If $\mathbb{E}_\mathbb{P}[h] = 1$ but $A_0 \neq \emptyset$ (i.e., $h = 1$ $\mathbb{P}$ -a.s. but not identically 1): the formulas give $\sup = 1$ and $\inf = \max(1 - \rho, 0)$. The infimum is achieved by the symmetric construction, moving mass to A_0 . $\square$

Lemma E.2 (Robust Expectation under ℓ_1 -Ball Perturbation for Binary Classifiers). *Let $h : \mathcal{X} \rightarrow \{0, 1\}$ be a binary classifier, and let $\{x_i\}_{i=1}^m \subseteq \mathcal{X}$ be a fixed set of m samples. Let $\hat{p}_m = (1/m, \dots, 1/m) \in \Delta^m$ denote the uniform empirical distribution, and for $\rho \geq 0$ define the uncertainty set*

$$\mathcal{U}_\rho = \{\tilde{p} \in \Delta^m : \|\tilde{p} - \hat{p}_m\|_1 \leq 2\rho\},$$

where $\Delta^m = \{\tilde{p} \in \mathbb{R}^m : \tilde{p}_i \geq 0, \sum_{i=1}^m \tilde{p}_i = 1\}$ is the probability simplex. The factor 2 ensures ρ corresponds to the total variation radius: $\text{TV}(\tilde{p}, \hat{p}_m) \leq \rho$. Let $\hat{\mathbb{E}}[h] = \frac{1}{m} \sum_{i=1}^m h(x_i)$ denote the empirical mean.

- If h is constant on the samples (i.e., $h(x_i) = c$ for all i with $c \in \{0, 1\}$), then

$$\sup_{\tilde{p} \in \mathcal{U}_\rho} \sum_{i=1}^m \tilde{p}_i h(x_i) = \inf_{\tilde{p} \in \mathcal{U}_\rho} \sum_{i=1}^m \tilde{p}_i h(x_i) = c.$$

- Otherwise, when h takes both values 0 and 1 on the samples:

$$\begin{aligned} \sup_{\tilde{p} \in \mathcal{U}_\rho} \sum_{i=1}^m \tilde{p}_i h(x_i) &= \min(\hat{\mathbb{E}}[h] + \rho, 1), \\ \inf_{\tilde{p} \in \mathcal{U}_\rho} \sum_{i=1}^m \tilde{p}_i h(x_i) &= \max(\hat{\mathbb{E}}[h] - \rho, 0). \end{aligned}$$

Proof of Lemma E.2. We prove the two cases separately.

Step 1: Setup and notation. Let $k = \sum_{i=1}^m h(x_i)$ denote the number of samples where h outputs 1. Define index sets $S_1 = \{i : h(x_i) = 1\}$ of size k and $S_0 = \{i : h(x_i) = 0\}$ of size $m - k$. The empirical mean is $\hat{\mathbb{E}}[h] = k/m$.

Any $\tilde{p} \in \mathcal{U}_\rho$ can be written as $\tilde{p} = \hat{p}_m + \delta$, where $\delta \in \mathbb{R}^m$ satisfies:

$$(C1) \sum_{i=1}^m \delta_i = 0 \quad (\text{since both } \tilde{p} \text{ and } \hat{p}_m \text{ sum to } 1), \quad (28)$$

$$(C2) \|\delta\|_1 = \sum_{i=1}^m |\delta_i| \leq 2\rho, \quad (29)$$

$$(C3) \delta_i \geq -1/m \text{ for all } i \quad (\text{to ensure } \tilde{p}_i = 1/m + \delta_i \geq 0). \quad (30)$$

Step 2: Constant classifier case ($k = 0$ or $k = m$). If $h(x_i) = c$ for all i (with $c \in \{0, 1\}$), then for any $\tilde{p} \in \Delta^m$,

$$\sum_{i=1}^m \tilde{p}_i h(x_i) = c \sum_{i=1}^m \tilde{p}_i = c \cdot 1 = c.$$

Since the objective is constant over all probability distributions, both the supremum and infimum over $\mathcal{U}_\rho \subseteq \Delta^m$ equal c .

Step 3: Non-constant case setup ($0 < k < m$). Since h takes both values on the samples, we have $k \in \{1, \dots, m-1\}$, so both S_0 and S_1 are nonempty and the divisions by k and $(m-k)$ below are well-defined.

Using $\tilde{p} = \hat{p}_m + \delta$, we reformulate the objective:

$$\sum_{i=1}^m \tilde{p}_i h(x_i) = \sum_{i=1}^m \left(\frac{1}{m} + \delta_i \right) h(x_i) = \frac{1}{m} \sum_{i=1}^m h(x_i) + \sum_{i=1}^m \delta_i h(x_i) = \hat{\mathbb{E}}[h] + \sum_{i \in S_1} \delta_i,$$

where the last equality uses $h(x_i) = 0$ for $i \in S_0$.

Step 4: Upper bound for the supremum. We show $\sum_{i \in S_1} \delta_i \leq \rho$ for any feasible δ .

Decompose $\delta_i = \delta_i^+ - \delta_i^-$ where $\delta_i^+ = \max(\delta_i, 0) \geq 0$ and $\delta_i^- = \max(-\delta_i, 0) \geq 0$. Constraints (28) and (29) together imply

$$\sum_{i=1}^m \delta_i^+ = \sum_{i=1}^m \delta_i^- = \frac{1}{2} \|\delta\|_1 \leq \rho.$$

Therefore,

$$\begin{aligned} \sum_{i \in S_1} \delta_i &= \sum_{i \in S_1} \delta_i^+ - \sum_{i \in S_1} \delta_i^- \\ &\leq \sum_{i \in S_1} \delta_i^+ \quad (\text{since } \delta_i^- \geq 0) \\ &\leq \sum_{i=1}^m \delta_i^+ \leq \rho. \end{aligned}$$

Thus, $\sum_{i=1}^m \tilde{p}_i h(x_i) \leq \hat{\mathbb{E}}[h] + \rho$. Additionally, since $h(x_i) \in \{0, 1\}$ and $\tilde{p} \in \Delta^m$, we have $\sum_{i=1}^m \tilde{p}_i h(x_i) \leq \sum_{i=1}^m \tilde{p}_i \cdot 1 = 1$.

Combining: $\sup_{\tilde{p} \in \mathcal{U}_\rho} \sum_{i=1}^m \tilde{p}_i h(x_i) \leq \min(\hat{\mathbb{E}}[h] + \rho, 1)$.

Step 5: Achieving the supremum. We construct feasible δ attaining the upper bound.

Case A. $\hat{\mathbb{E}}[h] + \rho \leq 1$ (equivalently, $\rho \leq (m - k)/m$).

Define

$$\delta_i = \begin{cases} \rho/k, & i \in S_1, \\ -\rho/(m - k), & i \in S_0. \end{cases}$$

Verification of constraints:

- (C1): $\sum_{i=1}^m \delta_i = k \cdot \frac{\rho}{k} + (m - k) \cdot \left(-\frac{\rho}{m-k}\right) = \rho - \rho = 0$. ✓
- (C2): $\|\delta\|_1 = k \cdot \frac{\rho}{k} + (m - k) \cdot \frac{\rho}{m-k} = \rho + \rho = 2\rho$. ✓
- (C3) for $i \in S_1$: $\delta_i = \rho/k > 0 \geq -1/m$. ✓
- (C3) for $i \in S_0$: $\delta_i = -\rho/(m - k) \geq -1/m$ iff $\rho \leq (m - k)/m$, which holds by assumption. ✓

Objective value:

$$\sum_{i=1}^m \tilde{p}_i h(x_i) = \hat{\mathbb{E}}[h] + \sum_{i \in S_1} \delta_i = \frac{k}{m} + k \cdot \frac{\rho}{k} = \frac{k}{m} + \rho = \hat{\mathbb{E}}[h] + \rho.$$

Case B. $\hat{\mathbb{E}}[h] + \rho > 1$ (equivalently, $\rho > (m - k)/m$).

Define

$$\delta_i = \begin{cases} \frac{m-k}{mk} = \frac{1}{k} - \frac{1}{m}, & i \in S_1, \\ -\frac{1}{m}, & i \in S_0. \end{cases}$$

Verification of constraints:

- (C1): $\sum_{i=1}^m \delta_i = k \cdot \frac{m-k}{mk} + (m - k) \cdot \left(-\frac{1}{m}\right) = \frac{m-k}{m} - \frac{m-k}{m} = 0$. ✓
- (C2): $\|\delta\|_1 = k \cdot \frac{m-k}{mk} + (m - k) \cdot \frac{1}{m} = \frac{m-k}{m} + \frac{m-k}{m} = \frac{2(m-k)}{m} \leq 2\rho$, where the inequality follows from the case condition $\rho > (m - k)/m$. ✓
- (C3) for $i \in S_1$: $\delta_i = (m - k)/(mk) = 1/k - 1/m \geq 0 > -1/m$ (since $k \leq m$). ✓
- (C3) for $i \in S_0$: $\delta_i = -1/m \geq -1/m$. ✓

Objective value:

$$\sum_{i=1}^m \tilde{p}_i h(x_i) = \sum_{i \in S_1} \left(\frac{1}{m} + \frac{m-k}{mk} \right) = k \cdot \left(\frac{1}{m} + \frac{m-k}{mk} \right) = \frac{k}{m} + \frac{m-k}{m} = 1.$$

In both cases, the supremum equals $\min(\hat{\mathbb{E}}[h] + \rho, 1)$.

Step 6: Infimum via duality. Define $g : \mathcal{X} \rightarrow \{0, 1\}$ by $g(x) = 1 - h(x)$. Then for any $\tilde{p} \in \Delta^m$,

$$\sum_{i=1}^m \tilde{p}_i h(x_i) = 1 - \sum_{i=1}^m \tilde{p}_i g(x_i).$$

Therefore,

$$\inf_{\tilde{p} \in \mathcal{U}_\rho} \sum_{i=1}^m \tilde{p}_i h(x_i) = 1 - \sup_{\tilde{p} \in \mathcal{U}_\rho} \sum_{i=1}^m \tilde{p}_i g(x_i).$$

Since h is non-constant, so is g . The empirical mean of g is $\hat{\mathbb{E}}[g] = 1 - \hat{\mathbb{E}}[h]$. Applying the supremum result (Steps 4–5) to g :

$$\sup_{\tilde{p} \in \mathcal{U}_\rho} \sum_{i=1}^m \tilde{p}_i g(x_i) = \min(\hat{\mathbb{E}}[g] + \rho, 1) = \min(1 - \hat{\mathbb{E}}[h] + \rho, 1).$$

Thus,

$$\inf_{\tilde{p} \in \mathcal{U}_\rho} \sum_{i=1}^m \tilde{p}_i h(x_i) = 1 - \min(1 - \hat{\mathbb{E}}[h] + \rho, 1).$$

We simplify as follows:

$$\inf_{\tilde{p} \in \mathcal{U}_\rho} \sum_{i=1}^m \tilde{p}_i h(x_i) = \begin{cases} \hat{\mathbb{E}}[h] - \rho, & \text{if } \rho \leq \hat{\mathbb{E}}[h], \\ 0, & \text{if } \rho > \hat{\mathbb{E}}[h], \end{cases} = \max(\hat{\mathbb{E}}[h] - \rho, 0).$$

Step 7: Conclusion. We have shown that for a constant classifier, the robust expectation equals the constant value c . For a non-constant classifier, the supremum is $\min(\hat{\mathbb{E}}[h] + \rho, 1)$ and the infimum is $\max(\hat{\mathbb{E}}[h] - \rho, 0)$. This completes the proof. $\square$

Lemma E.3 (Robust Expectation of Quadratic Functionals under TV Constraints). *Let $(\mathcal{X}, \mathcal{F})$ be a measurable space, $\mathbb{P}$ a probability distribution on $\mathcal{X}$, and $V : \mathcal{X} \times \mathcal{X} \rightarrow [0, 1]$ a measurable function. For any $\rho \geq 0$, define the Total-Variation ball on the product space:*

$$\mathcal{B}_\rho = \{Q : \text{TV}(Q, \mathbb{P} \otimes \mathbb{P}) \leq \rho\},$$

where Q is a distribution on $\mathcal{X} \times \mathcal{X}$. Define the functional:

$$\mathcal{Q}(Q) = \mathbb{E}_{(X, X') \sim Q}[V(X, X')].$$

Then for any $Q \in \mathcal{B}_\rho$:

$$|\mathcal{Q}(Q) - \mathcal{Q}(\mathbb{P} \otimes \mathbb{P})| \leq \rho.$$

Moreover, we have the one-sided bounds:

$$\sup_{Q \in \mathcal{B}_\rho} \mathcal{Q}(Q) \leq \min(\mathcal{Q}(\mathbb{P} \otimes \mathbb{P}) + \rho, 1) \quad \text{and} \quad \inf_{Q \in \mathcal{B}_\rho} \mathcal{Q}(Q) \geq \max(\mathcal{Q}(\mathbb{P} \otimes \mathbb{P}) - \rho, 0).$$

Proof. By the coupling characterization of total variation distance, for any Q with $\text{TV}(Q, \mathbb{P} \otimes \mathbb{P}) \leq \rho$ and any $\epsilon > 0$, there exists a coupling $((X, X'), (Y, Y'))$ such that:

$$(X, X') \sim \mathbb{P} \otimes \mathbb{P}, \quad (Y, Y') \sim Q, \quad \text{and} \quad \mathbb{P}((X, X') \neq (Y, Y')) \leq \rho + \epsilon.$$

Since V takes values in $[0, 1]$:

$$\begin{aligned} |\mathcal{Q}(Q) - \mathcal{Q}(\mathbb{P} \otimes \mathbb{P})| &= |\mathbb{E}[V(Y, Y')] - \mathbb{E}[V(X, X')]| \\ &\leq \mathbb{E}[|V(Y, Y') - V(X, X')|] \\ &\leq \mathbb{P}((Y, Y') \neq (X, X')) \leq \rho + \epsilon. \end{aligned}$$

Taking $\epsilon \rightarrow 0$ gives $|\mathcal{Q}(Q) - \mathcal{Q}(\mathbb{P} \otimes \mathbb{P})| \leq \rho$. Combined with $\mathcal{Q}(Q) \in [0, 1]$ for all Q , the one-sided bounds follow. $\square$

Lemma E.4 (Bounds for Robust Expectation of Bounded Functions (Discrete Case)). *Let $h : \mathcal{X} \rightarrow [0, 1]$ be a bounded function, and let $\{x_i\}_{i=1}^m \subseteq \mathcal{X}$ be a fixed set of m samples. Let $\hat{p}_m = (1/m, \dots, 1/m) \in \Delta^m$ denote the uniform empirical distribution, and for $\rho \geq 0$ define the uncertainty set*

$$\mathcal{U}_\rho = \{\tilde{p} \in \Delta^m : \|\tilde{p} - \hat{p}_m\|_1 \leq 2\rho\},$$

where $\Delta^m = \{\tilde{p} \in \mathbb{R}^m : \tilde{p}_i \geq 0, \sum_{i=1}^m \tilde{p}_i = 1\}$ is the probability simplex. (Note: $\|\tilde{p} - \hat{p}_m\|_1 = 2 \text{TV}(\tilde{p}, \hat{p}_m)$, so $\text{TV}(\tilde{p}, \hat{p}_m) \leq \rho$.) Let $\hat{\mathbb{E}}[h] = \frac{1}{m} \sum_{i=1}^m h(x_i)$ denote the empirical mean. Then

$$\begin{aligned} \sup_{\tilde{p} \in \mathcal{U}_\rho} \sum_{i=1}^m \tilde{p}_i h(x_i) &\leq \min(\hat{\mathbb{E}}[h] + \rho, 1), \\ \inf_{\tilde{p} \in \mathcal{U}_\rho} \sum_{i=1}^m \tilde{p}_i h(x_i) &\geq \max(\hat{\mathbb{E}}[h] - \rho, 0). \end{aligned}$$

Proof. Write $\tilde{p} = \hat{p}_m + \delta$, where $\delta \in \mathbb{R}^m$ satisfies:

- (i) $\sum_{i=1}^m \delta_i = 0$ (since both $\tilde{p}$ and $\hat{p}_m$ are probability vectors),
- (ii) $\|\delta\|_1 = \sum_{i=1}^m |\delta_i| \leq 2\rho$ (from the definition of $\mathcal{U}_\rho$),
- (iii) $\delta_i \geq -1/m$ for all i (since $\tilde{p}_i = 1/m + \delta_i \geq 0$ as $\tilde{p} \in \Delta^m$).

Decompose $\delta_i = \delta_i^+ - \delta_i^-$, where $\delta_i^+ = \max(\delta_i, 0) \geq 0$ and $\delta_i^- = \max(-\delta_i, 0) \geq 0$. From (i) we have $\sum_i \delta_i^+ = \sum_i \delta_i^-$. Combining with (ii) gives

$$\sum_{i=1}^m \delta_i^+ = \sum_{i=1}^m \delta_i^- = \frac{1}{2} \|\delta\|_1 \leq \rho.$$

Now,

$$\sum_{i=1}^m \tilde{p}_i h(x_i) = \hat{\mathbb{E}}[h] + \sum_{i=1}^m \delta_i h(x_i).$$

Supremum bound. Since $h(x_i) \in [0, 1]$,

$$\begin{aligned} \sum_{i=1}^m \delta_i h(x_i) &= \sum_{i=1}^m (\delta_i^+ - \delta_i^-) h(x_i) \\ &\leq \sum_{i=1}^m \delta_i^+ \cdot 1 - \sum_{i=1}^m \delta_i^- \cdot 0 \\ &= \sum_{i=1}^m \delta_i^+ \leq \rho. \end{aligned}$$

Thus, for any $\tilde{p} \in \mathcal{U}_\rho$,

$$\sum_{i=1}^m \tilde{p}_i h(x_i) \leq \hat{\mathbb{E}}[h] + \rho.$$

Also, because $h \leq 1$ and $\tilde{p}$ is a probability vector, $\sum_i \tilde{p}_i h(x_i) \leq 1$. Hence,

$$\sup_{\tilde{p} \in \mathcal{U}_\rho} \sum_{i=1}^m \tilde{p}_i h(x_i) \leq \min(\hat{\mathbb{E}}[h] + \rho, 1).$$

Infimum bound. Similarly,

$$\begin{aligned} \sum_{i=1}^m \delta_i h(x_i) &= \sum_{i=1}^m (\delta_i^+ - \delta_i^-) h(x_i) \\ &\geq \sum_{i=1}^m \delta_i^+ \cdot 0 - \sum_{i=1}^m \delta_i^- \cdot 1 \\ &= -\sum_{i=1}^m \delta_i^- \geq -\rho. \end{aligned}$$

Thus, for any $\tilde{p} \in \mathcal{U}_\rho$,

$$\sum_{i=1}^m \tilde{p}_i h(x_i) \geq \hat{\mathbb{E}}[h] - \rho.$$

Since $h \geq 0$, we also have $\sum_i \tilde{p}_i h(x_i) \geq 0$. Therefore,

$$\inf_{\tilde{p} \in \mathcal{U}_\rho} \sum_{i=1}^m \tilde{p}_i h(x_i) \geq \max(\hat{\mathbb{E}}[h] - \rho, 0).$$

□

E.1. Proof of Theorem 6.1

Proof. The proof proceeds in three steps: (1) bound the worst-case DP gap, (2) bound uniform convergence for robust individual fairness, and (3) combine all terms via standard ERM analysis.

Step 1: Bounding the worst-case DP gap. We establish uniform convergence of the empirical worst-case DP violation to its population counterpart.

Clean distribution containment. By Theorem 4.2, $\text{TV}(\mathbb{P}_{X,Y|A=j}, \mathbb{P}_{\text{pert},X,Y|A=j}) \leq \rho_{\text{DP},j}$, which ensures the clean distribution satisfies $\mathbb{P} \in \mathcal{B}_{\text{DP}}$, the uncertainty set centered at $\mathbb{P}_{\text{pert}}$. Hence any bound on worst-case fairness over $\mathcal{B}_{\text{DP}}$ applies to the clean population.

For group $j \in \{0, 1\}$, define:

$$\hat{\mathbb{E}}_j[h] = \frac{1}{n_j} \sum_{i \in S_j^c} h(x_i^c), \quad \mathbb{E}_{\text{pert},j}[h] = \mathbb{P}_{\text{pert}}(h(X) = 1 \mid A = j).$$

Define the approximation function $F : [0, 1]^2 \rightarrow [0, 1]$:

$$F(\mu_1, \mu_0) = \max\left\{\min(\mu_1 + \rho_1, 1) - \max(\mu_0 - \rho_0, 0), \min(\mu_0 + \rho_0, 1) - \max(\mu_1 - \rho_1, 0)\right\},$$

where $\rho_j = \rho_{\text{DP},j}$ as defined in Eq. (16). We require the following lemmas.

Lemma E.5 (Empirical Worst-Case Characterization). *For any $h : \mathcal{X} \rightarrow \{0, 1\}$, let $M(h) = \max_{(\tilde{p}_0, \tilde{p}_1) \in \mathcal{U}_{\text{DP}}} \hat{\Delta}_{\text{DP}}(h; S^c, \tilde{p}_0, \tilde{p}_1)$. Then $|M(h) - F(\hat{\mathbb{E}}_1[h], \hat{\mathbb{E}}_0[h])| \leq \rho_0$.*

Proof. Case 1: Non-constant classifier. If h takes both values in each group, Lemma E.2 gives:

$$\sup_{\tilde{p}_j \in \mathcal{U}_j} \sum_{i \in S_j^c} \tilde{p}_{j,i} h(x_i^c) = \min(\hat{\mathbb{E}}_j[h] + \rho_j, 1), \quad \inf_{\tilde{p}_j \in \mathcal{U}_j} \sum_{i \in S_j^c} \tilde{p}_{j,i} h(x_i^c) = \max(\hat{\mathbb{E}}_j[h] - \rho_j, 0).$$

Thus:

$$M(h) = \max\left\{\min(\hat{\mu}_1 + \rho_1, 1) - \max(\hat{\mu}_0 - \rho_0, 0), \min(\hat{\mu}_0 + \rho_0, 1) - \max(\hat{\mu}_1 - \rho_1, 0)\right\} = F(\hat{\mu}_1, \hat{\mu}_0).$$

Case 2: Constant classifier on both groups. If $h \equiv c_j$ on S_j^c for both j , then $\sup_{\tilde{p}_j} \sum \tilde{p}_{j,i} h(x_i^c) = \inf_{\tilde{p}_j} \sum \tilde{p}_{j,i} h(x_i^c) = c_j$ (since reweighting a constant yields the constant).

- $(c_0, c_1) = (0, 0)$: $M(h) = |0 - 0| = 0$. Meanwhile, $F(0, 0) = \max\{\min(\rho_1, 1) - \max(-\rho_0, 0), \min(\rho_0, 1) - \max(-\rho_1, 0)\} = \max\{\rho_1, \rho_0\} = \rho_0$ (since $\rho_0 \geq \rho_1$).
- $(c_0, c_1) = (1, 1)$: $M(h) = |1 - 1| = 0$. Similarly, $F(1, 1) = \max\{\min(1 + \rho_1, 1) - \max(1 - \rho_0, 0), \min(1 + \rho_0, 1) - \max(1 - \rho_1, 0)\} = \max\{1 - (1 - \rho_0), 1 - (1 - \rho_1)\} = \max\{\rho_0, \rho_1\} = \rho_0$.
- $(c_0, c_1) = (0, 1)$: $M(h) = |1 - 0| = 1$. And $F(1, 0) = \max\{\min(1 + \rho_1, 1) - \max(-\rho_0, 0), \min(\rho_0, 1) - \max(1 - \rho_1, 0)\} = \max\{1 - 0, \rho_0 - (1 - \rho_1)\} = \max\{1, \rho_0 + \rho_1 - 1\} = 1$ (since $\rho_0 + \rho_1 < 2$), so difference = 0.

- $(c_0, c_1) = (1, 0)$: By symmetry, $M(h) = 1$ and $F(0, 1) = 1$, difference = 0.

Case 3: Constant classifier on one group. If h constant only on group 0 with value $c_0 \in \{0, 1\}$ (group 1 non-constant):

$$M(h) = \max\left\{\min(\hat{\mu}_1 + \rho_1, 1) - c_0, c_0 - \max(\hat{\mu}_1 - \rho_1, 0)\right\}.$$

We compare with $F(\hat{\mu}_1, c_0)$, which uses $\min(c_0 + \rho_0, 1)$ and $\max(c_0 - \rho_0, 0)$ in place of c_0 :

- If $c_0 = 0$: $M(h)$ uses 0, while F uses $\max(-\rho_0, 0) = 0$ and $\min(\rho_0, 1) = \rho_0$. Difference $\leq \rho_0$.
- If $c_0 = 1$: $M(h)$ uses 1, while F uses $\min(1 + \rho_0, 1) = 1$ and $\max(1 - \rho_0, 0) = 1 - \rho_0$. Difference $\leq \rho_0$.

The symmetric case (group 1 constant, group 0 non-constant) yields difference $\leq \rho_1$ by the same argument.

Combining all cases: $|M(h) - F(\hat{\mu}_1, \hat{\mu}_0)| \leq \rho_0$. $\square$

Lemma E.6 (Population Worst-Case Characterization). *For any $h : \mathcal{X} \rightarrow \{0, 1\}$, let $P(h) = \sup_{Q \in \mathcal{B}_{\text{DP}}(\mathbb{P}_{\text{pert}})} \Delta_{\text{DP}}(h; Q)$. Then $|P(h) - F(\mathbb{E}_{\text{pert},1}[h], \mathbb{E}_{\text{pert},0}[h])| \leq \rho_0$.*

Proof. Identical structure to Lemma E.5, using Lemma E.1 (TV-ball version) instead of Lemma E.2. $\square$

Lemma E.7 (Lipschitz Property of F). *For all $\mu_1, \mu'_1, \mu_0, \mu'_0 \in [0, 1]$:*

$$|F(\mu_1, \mu_0) - F(\mu'_1, \mu'_0)| \leq |\mu_1 - \mu'_1| + |\mu_0 - \mu'_0|.$$

Proof. Each component $g_1(\mu_1, \mu_0) = \min(\mu_1 + \rho_1, 1) - \max(\mu_0 - \rho_0, 0)$ satisfies:

$$|g_1(\mu_1, \mu_0) - g_1(\mu'_1, \mu'_0)| \leq |\mu_1 - \mu'_1| + |\mu_0 - \mu'_0|$$

because $\min(\cdot, 1)$ and $\max(\cdot, 0)$ are 1-Lipschitz. Same for g_2 . The maximum of two 1-Lipschitz functions (in ℓ_1 sense) is 1-Lipschitz. $\square$

Uniform convergence of the DP gap. Let $\hat{\mu} = (\hat{\mathbb{E}}_1[h], \hat{\mathbb{E}}_0[h])$ and $\mu_{\text{pert}} = (\mathbb{E}_{\text{pert},1}[h], \mathbb{E}_{\text{pert},0}[h])$.

Step 1.1: Decomposition. By triangle inequality:

$$\begin{aligned} |M(h) - P(h)| &\leq |M(h) - F(\hat{\mu})| \\ &\quad + |F(\hat{\mu}) - F(\mu_{\text{pert}})| \\ &\quad + |F(\mu_{\text{pert}}) - P(h)|. \end{aligned}$$

Step 1.2: Apply characterization lemmas. From Lemmas E.5 and E.6:

$$|M(h) - F(\hat{\mu})| \leq \rho_0, \quad |F(\mu_{\text{pert}}) - P(h)| \leq \rho_0.$$

From Lemma E.7:

$$|F(\hat{\mu}) - F(\mu_{\text{pert}})| \leq |\hat{\mathbb{E}}_1[h] - \mathbb{E}_{\text{pert},1}[h]| + |\hat{\mathbb{E}}_0[h] - \mathbb{E}_{\text{pert},0}[h]|.$$

Thus:

$$|M(h) - P(h)| \leq 2\rho_0 + |\hat{\mathbb{E}}_1[h] - \mathbb{E}_{\text{pert},1}[h]| + |\hat{\mathbb{E}}_0[h] - \mathbb{E}_{\text{pert},0}[h]|.$$

Step 1.3: Uniform convergence of acceptance rates. The group-conditional empirical means $\hat{\mathbb{E}}_j[h]$ converge uniformly to their population counterparts $\mathbb{E}_{\text{pert},j}[h]$. By the same machinery as Steps 4a–4c and 5 in the proof of Lemma D.3 (Chernoff concentration for group sizes ensuring $n_j \geq (1 - \alpha)\pi_j n/2$ under the sample requirement (21), followed by VC uniform convergence (Theorem 3.3) on group- j samples, and a union bound over both groups), we obtain with probability at least $1 - \delta/3$:

$$B_j \leq 2\sqrt{2} \sqrt{\frac{8D_{\text{VC}} \log(en/D_{\text{VC}}) + 2\log(48/\delta)}{(1 - \alpha)\pi_j n}}.$$

Since $\pi_0 \leq \pi_1$ (Section 3.1), we have $B_0 \geq B_1$, so $B_0 + B_1 \leq 2B_0$. Combining with Step 1.2:

$$\sup_{h \in \mathcal{H}} |M(h) - P(h)| \leq \underbrace{2\rho_0}_{\text{char. error}} + \underbrace{4\sqrt{2} \sqrt{\frac{8D_{\text{VC}} \log(en/D_{\text{VC}}) + 2\log(48/\delta)}{(1-\alpha)\pi_0 n}}}_{\text{sampling error}}. \quad (31)$$

Step 2: Uniform convergence for robust individual fairness. We now bound the uniform deviation for the robust individual fairness term. Starting from the weighted empirical IF loss (Eq. (15)):

$$\widehat{L}_{d,\gamma}(h; S^c, \tilde{p}) = \frac{1}{n-1} \sum_{i \neq j} \frac{\tilde{p}_i + \tilde{p}_j}{2} \cdot \mathbb{1}(|h(x_i^c) - h(x_j^c)| > d(x_i^c, x_j^c) + \gamma).$$

Define the pairwise violation indicator $V_{ij}(h) := \mathbb{1}(|h(x_i^c) - h(x_j^c)| > d(x_i^c, x_j^c) + \gamma) \in \{0, 1\}$. Since the distance d is symmetric (Definition 3.2), we have $V_{ij} = V_{ji}$. By rearranging:

$$\widehat{L}_{d,\gamma}(h; \tilde{p}) = \frac{1}{n-1} \sum_{i \neq j} \frac{\tilde{p}_i + \tilde{p}_j}{2} V_{ij} = \sum_{i=1}^n \tilde{p}_i \cdot \underbrace{\frac{1}{n-1} \sum_{j \neq i} V_{ij}}_{=: \bar{\ell}_i(h)} = \langle \tilde{p}, \bar{\ell}(h) \rangle, \quad (32)$$

where $\bar{\ell}_i(h) \in [0, 1]$ since $\sum_{j \neq i} V_{ij} \leq n-1$.

Let $\mathcal{U}_{\text{IF}} = \{\tilde{p} \in \Delta^n : \|\tilde{p} - \hat{p}_n\|_1 \leq 2\rho_{\text{IF}}\}$.

Robust expectation bounds. Applying Lemma E.4 to the empirical case:

$$\max_{\tilde{p} \in \mathcal{U}_{\text{IF}}} \widehat{L}_{d,\gamma}(h; \tilde{p}) \leq \min(\widehat{L}_{d,\gamma}(h; \hat{p}_n) + \rho_{\text{IF}}, 1).$$

For the population case, define $V(x, x') = \mathbb{1}(|h(x) - h(x')| > d(x, x') + \gamma) \in [0, 1]$. Then $L_{d,\gamma}(h; Q) = \mathbb{E}_{(X, X') \sim Q}[V(X, X')]$ for any distribution Q on $\mathcal{X} \times \mathcal{X}$. Define the uncertainty set on the product space: $\mathcal{B}_{\text{IF}} = \{Q : \text{TV}(Q, \mathbb{P}_{\text{pert}, X}^{\otimes 2}) \leq \rho_{\text{IF}}\}$. Applying Lemma E.3:

$$\sup_{Q \in \mathcal{B}_{\text{IF}}} L_{d,\gamma}(h; Q) \leq \min(L_{d,\gamma}(h; \mathbb{P}_{\text{pert}}) + \rho_{\text{IF}}, 1).$$

Bounding the deviation. Since $\hat{p}_n \in \mathcal{U}_{\text{IF}}$ (zero perturbation) and $\mathbb{P}_{\text{pert}, X}^{\otimes 2} \in \mathcal{B}_{\text{IF}}$ (center of the TV ball):

$$\begin{aligned} \widehat{L}_{d,\gamma}(h; \hat{p}_n) &\leq \max_{\tilde{p} \in \mathcal{U}_{\text{IF}}} \widehat{L}_{d,\gamma}(h; \tilde{p}) \leq \widehat{L}_{d,\gamma}(h; \hat{p}_n) + \rho_{\text{IF}}, \\ L_{d,\gamma}(h; \mathbb{P}_{\text{pert}}) &\leq \sup_{Q \in \mathcal{B}_{\text{IF}}} L_{d,\gamma}(h; Q) \leq L_{d,\gamma}(h; \mathbb{P}_{\text{pert}}) + \rho_{\text{IF}}. \end{aligned}$$

Thus, by the triangle inequality:

$$\begin{aligned} &\left| \max_{\tilde{p} \in \mathcal{U}_{\text{IF}}} \widehat{L}_{d,\gamma}(h; \tilde{p}) - \sup_{\mu \in \mathcal{B}_{\text{IF}}} L_{d,\gamma}(h; \mu) \right| \\ &\leq \left| \widehat{L}_{d,\gamma}(h; \hat{p}_n) - L_{d,\gamma}(h; \mathbb{P}_{\text{pert}}) \right| + \rho_{\text{IF}}. \end{aligned}$$

Sampling error bound. By the same argument as in the proof of Lemma D.2 (which establishes that the pairwise violation class has VC dimension at most $2D_{\text{VC}}$), applying U-process concentration (Nolan & Pollard, 1987; Cl  men  on et al., 2008), with probability at least $1 - \delta/3$:

$$\sup_{h \in \mathcal{H}} \left| \max_{\tilde{p} \in \mathcal{U}_{\text{IF}}} \widehat{L}_{d,\gamma}(h; \tilde{p}) - \sup_{Q \in \mathcal{B}_{\text{IF}}} L_{d,\gamma}(h; Q) \right| \leq \rho_{\text{IF}} + 2\sqrt{\frac{16D_{\text{VC}} \log(en/(2D_{\text{VC}})) + 2\log(12/\delta)}{n}}. \quad (33)$$

Step 3: Combining via ERM analysis. Let $h^* = \arg \min_{h \in \mathcal{H}} \mathcal{L}_{\text{DRO}}(h; \mathbb{P}, \mathcal{B})$ denote the population DRO minimizer. By standard ERM generalization analysis, since $\hat{h}$ minimizes the empirical DRO objective, the excess population risk satisfies:

$$\mathcal{L}_{\text{DRO}}(\hat{h}; \mathbb{P}, \mathcal{B}) - \mathcal{L}_{\text{DRO}}(h^*; \mathbb{P}, \mathcal{B}) \leq 2 \sup_{h \in \mathcal{H}} \left| \hat{\mathcal{L}}_{\text{DRO}}(h; S^c, \mathcal{U}) - \mathcal{L}_{\text{DRO}}(h; \mathbb{P}, \mathcal{B}) \right|. \quad (34)$$

Risk term analysis. By Lemma D.1, with probability at least $1 - \delta/3$,

$$\sup_{h \in \mathcal{H}} |R(h; \mathbb{P}) - \hat{R}(h; S^c)| \leq \frac{3\alpha}{2} + 2\sqrt{\frac{8D_{\text{VC}} \log(en/D_{\text{VC}}) + 2\log(24/\delta)}{n}}.$$

DP term analysis. From Step 1 (Eq. (31)), with probability at least $1 - \delta/3$,

$$\sup_{h \in \mathcal{H}} \left| \max_{\tilde{p} \in \mathcal{U}_{\text{DP}}} \hat{\Delta}_{\text{DP}}(h; \tilde{p}) - \sup_{Q \in \mathcal{B}_{\text{DP}}} \Delta_{\text{DP}}(h; Q) \right| \leq 2\rho_0 + 4\sqrt{2} \sqrt{\frac{8D_{\text{VC}} \log(en/D_{\text{VC}}) + 2\log(48/\delta)}{(1-\alpha)\pi_0 n}}.$$

IF term analysis. From Step 2 (Eq. (33)), with probability at least $1 - \delta/3$,

$$\sup_{h \in \mathcal{H}} \left| \max_{\tilde{p} \in \mathcal{U}_{\text{IF}}} \hat{L}_{d,\gamma}(h; \tilde{p}) - \sup_{Q \in \mathcal{B}_{\text{IF}}} L_{d,\gamma}(h; Q) \right| \leq \rho_{\text{IF}} + 2\sqrt{\frac{16D_{\text{VC}} \log(en/(2D_{\text{VC}})) + 2\log(12/\delta)}{n}}.$$

Combining via union bound. Taking a union bound over the three events (each with failure probability $\delta/3$), all bounds hold simultaneously with probability at least $1 - \delta$. The excess risk (34) satisfies:

$$\begin{aligned} \sup_{h \in \mathcal{H}} |\mathcal{L}_{\text{DRO}}(h) - \hat{\mathcal{L}}_{\text{DRO}}(h)| &\leq \left[\frac{3\alpha}{2} + 2\sqrt{\frac{8D_{\text{VC}} \log\left(\frac{en}{D_{\text{VC}}}\right) + 2\log\left(\frac{24}{\delta}\right)}{n}} \right] \\ &\quad + \lambda_{\text{IF}} \left[\rho_{\text{IF}} + 2\sqrt{\frac{16D_{\text{VC}} \log(en/(2D_{\text{VC}})) + 2\log(12/\delta)}{n}} \right] \\ &\quad + \lambda_{\text{DP}} \left[2\rho_0 + 4\sqrt{2} \sqrt{\frac{8D_{\text{VC}} \log\left(\frac{en}{D_{\text{VC}}}\right) + 2\log\left(\frac{48}{\delta}\right)}{(1-\alpha)\pi_0 n}} \right]. \end{aligned}$$

By the standard ERM symmetrization argument (34), doubling this bound yields the excess risk Δ_{DRO} .

This completes the proof. $\square$

F. Datasets and Random vs. Adversarial Corruption

Our theoretical bounds (Section 4) hold for *worst-case* adversarial corruption within the α -fraction budget: an adversary may arbitrarily modify up to αn samples to maximally degrade fairness. In contrast, the experimental corruption protocol (Section 7) is *random*: samples are selected uniformly with probability α , and corruptions are applied stochastically (Gaussian noise for numeric features, uniform random replacement for categorical features, label flipping, and protected attribute flipping).

Implications. Random corruption models realistic data quality issues (measurement errors, data entry mistakes, and collection biases) that arise without adversarial intent. Because our theoretical bounds hold for worst-case corruption, they remain valid under random corruption, which is strictly less adversarial. Consequently, the experimental results provide a *lower bound* on the degradation that would occur under strategic adversarial corruption. Under such attacks, we expect Naive-FAIR to degrade further while DRO-FAIR maintains its robustness guarantees by design.

Group Proportion Estimation. In our experiments, we consider a corruption model where protected attributes are flipped independently with probability α . Under this model, the empirical proportion $\hat{\pi}_j^{\text{obs}} = n_j/n$ from S^c relates to the clean proportion π_j as $\hat{\pi}_j^{\text{obs}} = (1 - \alpha)\pi_j + \alpha(1 - \pi_j)$. We estimate π_j via:

$$\hat{\pi}_j = \frac{\hat{\pi}_j^{\text{obs}} - \alpha}{1 - 2\alpha} \quad \text{for } \alpha \neq 0.5.$$

This bias-corrected estimate ensures the radius $\rho_{\text{DP},j}$ in (16) accurately reflects the distribution shift. For $\alpha = 0.5$, the model is non-identifiable; we use $\hat{\pi}_j^{\text{obs}}$ as a conservative approximation.

F.1. Baseline Selection Rationale

Naive-FAIR is the natural ablation that isolates the contribution of robust optimization. Existing group-DRO methods (Hashimoto et al., 2018; Sagawa et al., 2020) address *natural* distribution shift over clean subpopulations, not adversarial α -corruption of features, labels, and protected attributes. While Blum et al. (2024) analyze label-flipping attacks, they establish hardness results rather than constructive algorithms and do not address individual fairness. Wang et al. (2020) address noisy protected group labels via robust optimization; their setting is a special case of our corruption model when individual fairness is disabled (see Appendix 7.4 for a DP-only ablation). However, they assume statistical noise rather than adversarial corruption. Adapting these methods to our full corruption model would require non-trivial modifications that are orthogonal to our contribution. A comprehensive comparison across these paradigms is an important direction for future work.

G. DRO-FAIR Algorithm Details

This appendix provides the complete empirical formulation and optimization algorithm for the DRO-FAIR framework introduced in Section 6.

G.1. Notation and Preliminaries

We use notation established in Section 6.1. For convenience, we restate: $S^c = \{(x_i^c, a_i^c, y_i^c)\}_{i=1}^n$ is the corrupted training sample, Δ^m the m -simplex, and $\hat{p}_m = (1/m, \dots, 1/m)$ the uniform distribution. For each group $j \in \{0, 1\}$, let $n_j = |\{i : a_i^c = j\}|$ denote the group size.

For differentiable optimization, we use a soft classifier $\tilde{h}_\theta(x) = \sigma(\tau \cdot f_\theta(x)) \in [0, 1]$, where $f_\theta : \mathcal{X} \rightarrow \mathbb{R}$ is a neural network and σ is the sigmoid function; see Section G.6 for details on surrogate losses and test-time evaluation. We use $d : \mathcal{X} \times \mathcal{X} \rightarrow [0, 1]$ for the similarity metric and $\gamma \geq 0$ for the fairness margin (Definition 3.2).

G.2. Empirical Lagrangian Formulation

The constrained optimization problem (17) from Section 6.1 is reformulated as an unconstrained Lagrangian min-max problem. Extending the formulation (18) with a tilted empirical risk (which approximates CVaR) and replacing the 0-1 loss with a smooth convex surrogate $\ell : \mathbb{R} \times \{0, 1\} \rightarrow [0, 1]$ (e.g., binary cross-entropy), we obtain:

$$\begin{aligned} \min_{\theta} \max_{\lambda_{\text{DP}}, \lambda_{\text{IF}} \geq 0} & \left\{ \underbrace{\beta \log \left(\frac{1}{n} \sum_{i=1}^n \exp \left(\frac{\ell(f_\theta(x_i^c), y_i^c)}{\beta} \right)} \right)}_{\text{Tilted empirical risk (approximates CVaR)}} \right. \\ & + \lambda_{\text{DP}} \cdot \max_{(\tilde{p}_0, \tilde{p}_1) \in \mathcal{U}_{\text{DP}}} g_{\text{DP}}(\theta; \tilde{p}_0, \tilde{p}_1) \\ & \left. + \lambda_{\text{IF}} \cdot \max_{\tilde{p} \in \mathcal{U}_{\text{IF}}} g_{\text{IF}}(\theta; \tilde{p}) \right\}, \end{aligned} \quad (35)$$

where $\beta > 0$ controls the tilting (recovering ERM as $\beta \rightarrow \infty$, worst-case risk as $\beta \rightarrow 0$), and $\lambda_{\text{DP}}, \lambda_{\text{IF}} \geq 0$ are Lagrange multipliers updated via dual ascent. The fairness violation functions g_{DP} and g_{IF} are defined explicitly in Section G.3 below. In our implementation, fairness tolerance thresholds are set to zero, enforcing strict constraint satisfaction.

G.3. Explicit Fairness Violation Functions

For notational compactness, we define $g_{\text{DP}} \equiv \hat{\Delta}_{\text{DP}}$ and $g_{\text{IF}} \equiv \hat{L}_{d,\gamma}$ (cf. Eqs. (14)–(15)). The weighted demographic parity violation is:

$$g_{\text{DP}}(\theta; \tilde{p}_0, \tilde{p}_1) = \left| \sum_{i: a_i^c=1} \tilde{p}_{1,i} \tilde{h}_\theta(x_i^c) - \sum_{i: a_i^c=0} \tilde{p}_{0,i} \tilde{h}_\theta(x_i^c) \right|, \quad (36)$$

where $\tilde{p}_{j,i}$ denotes the weight for sample i within group j , satisfying $\tilde{p}_{j,i} \geq 0$ and $\sum_{i: a_i^c=j} \tilde{p}_{j,i} = 1$, and $\tilde{h}_\theta = \sigma(\tau \cdot f_\theta)$ is the soft classifier (cf. Section G.1). This measures the absolute difference in weighted positive classification rates between groups.

The weighted individual fairness violation is computed over a k -nearest neighbor graph, where $\mathcal{N}(i)$ denotes the set of k nearest neighbors of x_i^c in the corrupted sample S^c :

$$g_{\text{IF}}(\theta; \tilde{p}) = \frac{1}{n-1} \sum_{i=1}^n \sum_{j \in \mathcal{N}(i)} \frac{\tilde{p}_i + \tilde{p}_j}{2} \cdot (|\tilde{h}_\theta(x_i^c) - \tilde{h}_\theta(x_j^c)| - d(x_i^c, x_j^c) - \gamma)_+, \quad (37)$$

where $(x)_+ := \max(0, x)$ denotes the positive part (hinge function) and $\tilde{p} \in \Delta^n$. This function acts as a convex relaxation of the 0-1 indicator from Eq. (15), quantifying the magnitude of violation for pairs that fail the fairness constraint while ignoring satisfied pairs.

Remark G.1 (Factored IF Uncertainty Set and k-NN Approximation). The pairwise weight $(\tilde{p}_i + \tilde{p}_j)/2$ in Eq. (37) corresponds to a *factored uncertainty set* where each sample has a single weight. A fully general uncertainty set over pairs would require $O(n^2)$ weight parameters; our factored parameterization requires only $O(n)$ parameters while still providing a valid robustness certificate.

k-NN Approximation. We evaluate IF over a k -nearest neighbor graph ($O(kn)$ pairs) rather than all $O(n^2)$ pairs. Since k-NN neighbors are the most similar pairs and thus most likely to violate IF constraints, this captures the relevant violations efficiently. We use $k = 5$ in all experiments.

G.4. Algorithm

Algorithm 1 presents the complete optimization procedure. The algorithm alternates between: (1) outer minimization to update model parameters, (2) dual ascent to update Lagrange multipliers, and (3) inner maximization to find worst-case distributions within the uncertainty sets.

Remark G.2 (Naive Baseline). Setting $\rho_{\text{DP},j} = \rho_{\text{IF}} = 0$ and removing the inner maximization loop in Algorithm 1 recovers the Naive-FAIR baseline. This corresponds to using fixed uniform weights $\tilde{p}_j = \hat{p}_{n,j}$ and $\tilde{p}^{\text{IF}} = \hat{p}_n$ throughout training, without robust reweighting. The Naive-FAIR approach is thus a special case that ignores worst-case distributional robustness.

G.5. Implementation Details

All datasets are preprocessed with label encoding for categorical variables and StandardScaler normalization. We use an 80/20 train-test split, with training data further split 85/15 for training/validation.

Inner Maximization. With classifier predictions held fixed, both g_{DP} and g_{IF} are piecewise-linear in the importance weights. We compute gradients via automatic differentiation and apply projected gradient ascent onto the uncertainty set after each update.

Projection onto ℓ_1 -Ball Intersected with Simplex. The uncertainty sets require projecting onto $\{p \in \Delta^m : \|p - \hat{p}\|_1 \leq 2\rho\}$, the intersection of the probability simplex and an ℓ_1 -ball centered at the uniform distribution. We use Dykstra’s alternating projection algorithm (Boyle & Dykstra, 1986), which iterates between simplex projection and ℓ_1 -ball projection until convergence.

Numerical Stability. The tilted objective is implemented using a numerically stable softmax-weighted formulation: $L_{\text{tilt}} = \sum_{i=1}^n \omega_i \ell(f_\theta(x_i^c), y_i^c)$, where $\omega = \text{softmax}((\ell - \ell_{\max})/\beta)$ with $\ell_{\max}$ subtracted before exponentiation to prevent overflow. Gradients flow through the softmax weights ω_i , providing implicit regularization for training stability.

Algorithm 1 Robust Fair Classification via DRO (DROF)

Require: Dataset $S^c = \{(x_i^c, a_i^c, y_i^c)\}_{i=1}^n$; radii $\rho_{\text{DP},j}, \rho_{\text{IF}}$; learning rates $\eta_\theta, \eta_\lambda, \eta_p$; epochs T ; inner steps K ; tilting β ; temperature τ ; neighbors k

- 1: **Initialize:** $\theta \leftarrow \theta_0, \lambda_{\text{DP}}, \lambda_{\text{IF}} \leftarrow \lambda_0$
- 2: **Initialize weights:** $\tilde{p}_j \leftarrow \hat{p}_{n_j}$ for $j \in \{0, 1\}$; $\tilde{p}^{\text{IF}} \leftarrow \hat{p}_n$
- 3: **Precompute:** k -NN graph $\mathcal{N}(i)$ for all $i \in \{1, \dots, n\}$
- 4: **for** $t = 1$ **to** T **do**
- 5: **for** each minibatch $\mathcal{M} \subset \{1, \dots, n\}$ **do**
- 6: *// Forward pass*
- 7: $\tilde{h}_i \leftarrow \sigma(\tau \cdot f_\theta(x_i^c))$ for $i \in \mathcal{M}$ {soft predictions}
- 8: *// Compute losses*
- 9: $L_{\text{tilt}} \leftarrow \beta \log \left(\frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \exp(\ell(f_\theta(x_i^c), y_i^c)/\beta) \right)$
- 10: $\bar{h}_j \leftarrow \sum_{i: a_i^c = j} \tilde{p}_{j,i} \tilde{h}_i$ for $j \in \{0, 1\}$ {weighted group rates}
- 11: $g_{\text{DP}} \leftarrow |\bar{h}_1 - \bar{h}_0|$
- 12: $g_{\text{IF}} \leftarrow \frac{1}{n-1} \sum_{i=1}^n \sum_{j \in \mathcal{N}(i)} \frac{\tilde{p}_i^{\text{IF}} + \tilde{p}_j^{\text{IF}}}{2} (|\tilde{h}_i - \tilde{h}_j| - d_{ij} - \gamma)_+$
- 13: *// Outer minimization: update model*
- 14: $\mathcal{L} \leftarrow L_{\text{tilt}} + \lambda_{\text{DP}} g_{\text{DP}} + \lambda_{\text{IF}} g_{\text{IF}}$
- 15: $\theta \leftarrow \theta - \eta_\theta \nabla_\theta \mathcal{L}$
- 16: *// Dual ascent: update multipliers*
- 17: $\lambda_{\text{DP}} \leftarrow \max(0, \lambda_{\text{DP}} + \eta_\lambda g_{\text{DP}})$
- 18: $\lambda_{\text{IF}} \leftarrow \max(0, \lambda_{\text{IF}} + \eta_\lambda g_{\text{IF}})$
- 19: *// Inner maximization: projected gradient ascent on importance weights*
- 20: **for** $s = 1$ **to** K **do**
- 21: $\tilde{p}_0 \leftarrow \text{Proj}_{\mathcal{U}_0}(\tilde{p}_0 + \eta_p \nabla_{\tilde{p}_0} g_{\text{DP}})$ {group 0 weights}
- 22: $\tilde{p}_1 \leftarrow \text{Proj}_{\mathcal{U}_1}(\tilde{p}_1 + \eta_p \nabla_{\tilde{p}_1} g_{\text{DP}})$ {group 1 weights}
- 23: $\tilde{p}^{\text{IF}} \leftarrow \text{Proj}_{\mathcal{U}^{\text{IF}}}(\tilde{p}^{\text{IF}} + \eta_p \nabla_{\tilde{p}^{\text{IF}}} g_{\text{IF}})$ {IF weights}
- 24: **end for**
- 25: **end for**
- 26: **end for**
- 27: **return** θ

Optimization. We use AdamW (Loshchilov & Hutter, 2019) for θ -updates with gradient clipping, Adam for λ -updates, and Adam for $\tilde{p}$ -updates. We clamp $\lambda \in [0, \lambda_{\text{max}}]$ for stability. We set $K = 10$ inner steps, $k = 5$ for the k -NN graph, $\beta = 5$ for tilting, and $\tau = 100$ for temperature.

Hyperparameter Sensitivity. The temperature τ should be tuned based on corruption level: $\tau = 100$ works well for moderate corruption ($\alpha \leq 0.3$), while reducing to $\tau = 1$ at $\alpha = 0.4$ on Adult improves both accuracy (79.6% vs 75.3%) and fairness (95% lower DP, 68% lower IF versus Naive-FAIR).

G.6. Bridging Theory and Practice: Surrogate Losses

Our theoretical analysis assumes binary classifiers with 0-1 loss. In practice, we use differentiable surrogates for gradient-based optimization.

Classification Loss. We minimize binary cross-entropy (BCE) on raw logits $f_\theta(x)$, a calibrated surrogate for 0-1 loss (Bartlett et al., 2006).

Fairness Losses. For computing fairness constraints during training, we use temperature-scaled predictions $\tilde{h}_\theta(x) = \sigma(\tau \cdot f_\theta(x))$ with $\tau = 100$ to produce near-binary outputs. For DP, we compute $|\mathbb{E}[\tilde{h}_\theta|A = 1] - \mathbb{E}[\tilde{h}_\theta|A = 0]|$, which approximates the binary DP violation. For IF, we use a hinge loss $(|\tilde{h}_i - \tilde{h}_j| - d_{ij} - \gamma)_+$, which approximates the binary IF violation.

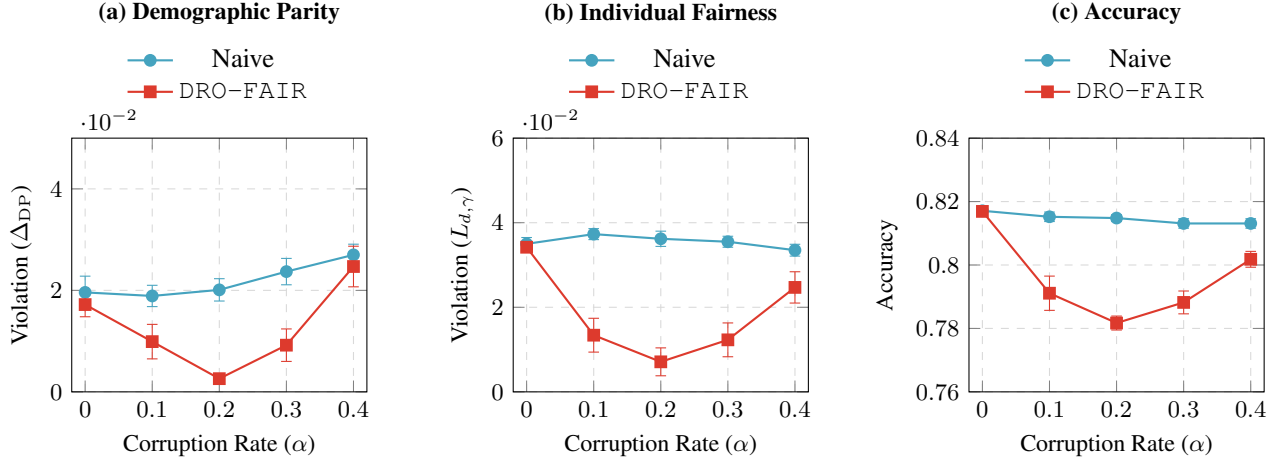


Figure 1. **Credit Card Default Results.** DRO-FAIR achieves consistently lower fairness violations across corruption rates, with 2 to 2.5 $\times$ IF improvement at moderate α and minimal accuracy cost ($\sim 2\%$).

Table 7. Similarity between learned IF and DP weights by subgroup.

Subgroup	Spearman R	Cosine Sim.
Global (All Data)	0.0127	0.1606
Female (Unpriv.), Pos	0.0157	0.0539
Male (Priv.), Neg	0.2064	0.2196
Male (Priv.), Pos	0.1206	0.1759
Female (Unpriv.), Neg	0.0023	0.0779

Test-Time Evaluation. All reported metrics are computed using standard sigmoid predictions thresholded at 0.5: $h(x) = \mathbb{1}[\sigma(f_\theta(x)) \geq 0.5]$, ensuring theoretical guarantees apply.

H. Analysis of Learned Weights

To better understand how DRO-FAIR achieves robustness, we analyze the importance weights learned during optimization on the Credit dataset with $\alpha = 0.2$. The re-weighting mechanism upweights “hard” samples to identify the worst-case distribution within the uncertainty set. For interpretability, we report **normalized importance weights** centered at 1.0: for IF, we normalize globally as $\bar{w}_i^{\text{IF}} = n \cdot \tilde{p}_i$; for DP, we normalize within each group as $\bar{w}_i^{\text{DP}} = n_j \cdot \tilde{p}_{j,i}$ for sample i in group j . A weight $\bar{w}_i > 1$ indicates a sample is upweighted relative to the uniform distribution.

Weight Similarity. We first examine whether the optimization identifies similar samples as “hard” for both Individual Fairness (IF) and Demographic Parity (DP). Table 2 presents the Spearman correlation and Cosine similarity between the learned IF and DP weights. The low global correlation (0.0127) suggests that the two fairness notions prioritize different subsets of the data, highlighting the importance of the joint optimization in Eq. (19).

Subgroup Analysis. We further aggregate the weights by subgroup (Table 3). Key observations:

- For **Individual Fairness**, the optimization assigns the highest weights to *Male, Pos* (1.27) and *Female, Pos* (1.25).
- For **Demographic Parity**, the weights focus on *Male, Pos* (1.27) and *Female, Neg* (1.07).

The disparity in weight distribution is most pronounced for the *Female, Pos* subgroup, with a mean per-sample absolute difference of 1.43.

Table 8. Mean learned importance weights by subgroup. The last column shows the mean per-sample absolute difference $\mathbb{E}[|\bar{w}_i^{\text{DP}} - \bar{w}_i^{\text{IF}}|]$.

Subgroup	IF Weight	DP Weight	Mean $ \Delta\bar{w} $
Female, Pos	1.25 ± 0.20	0.76 ± 0.07	1.43
Male, Neg	0.90 ± 0.06	0.86 ± 0.03	0.46
Male, Pos	1.27 ± 0.14	1.27 ± 0.07	1.19
Female, Neg	0.86 ± 0.07	1.07 ± 0.04	0.92

I. DP-IF Constraint Ablation

To investigate the DP-IF trade-off observed in Table 1, we conduct an ablation study removing the IF constraint from both DRO-FAIR and Naive-FAIR on LSAC at $\alpha = 0.4$. Table 4 presents the results.

Table 9. Ablation: Effect of removing IF constraint on LSAC ($\alpha = 0.4$).

Method	Acc $\uparrow$	DP $\downarrow$	IF $\downarrow$
Naive-FAIR (with IF)	.896	.009	.017
Naive-FAIR (no IF)	.897	.009	.025
DRO-FAIR (with IF)	.888	.029	.011
DRO-FAIR (no IF)	.897	.020	.027

Key findings: Removing the IF constraint improves DRO-FAIR’s DP from 0.029 to 0.020 and accuracy from 0.888 to 0.897, confirming that the IF constraint competes with DP optimization. This explains the DP-IF trade-off observed in the main results for LSAC at higher corruption levels. When constraints compete, practitioners may need to prioritize one fairness notion or adjust the Lagrange multiplier weights $\lambda_{\text{DP}}, \lambda_{\text{IF}}$.

J. Radius Selection Ablation

We empirically validate the importance of using calibrated radii (Remark 6.2) by comparing DRO-FAIR with calibrated radii versus overly large radii ($\rho_{\text{DP}} = \rho_{\text{IF}} = 1$). Table 5 presents results on the Credit dataset.

Table 10. Ablation: Effect of radius selection on Credit dataset.

α	Radii	Acc	DP $\downarrow$	IF $\downarrow$
0.1	Calibrated	.791	.010	.013
	$\rho_{\text{DP},j} = \rho_{\text{IF}} = 1$	.789	.016	.040
0.2	Calibrated	.782	.003	.007
	$\rho_{\text{DP},j} = \rho_{\text{IF}} = 1$	.808	.025	.025

Key findings: Using overly large radii ($\rho_{\text{DP},j} = \rho_{\text{IF}} = 1$) makes the worst-case constraints so conservative that the optimization struggles to satisfy them, effectively ignoring the fairness constraints. This results in (1) higher accuracy (constraints are not enforced) and (2) substantially worse fairness: DP violations are $2.5\times$ to $8\times$ higher, and IF violations are $3\times$ higher compared to calibrated radii. Conversely, as $\rho \rightarrow 0$, the uncertainty sets shrink to point masses at the empirical distribution, and DRO-FAIR recovers the Naive-FAIR approach (no robustness). The calibrated radii (Theorems 4.2–4.3) balance this trade-off: they are large enough to contain the clean distribution $\mathbb{P}$ within the uncertainty set, yet small enough to yield meaningful worst-case constraints.